The Pythagorean Three-Body Conjecture

September 6, 2026

Abstract

We study collision-free periodicity of Newtonian free-fall solutions whose masses equal the opposite sides of an integer right triangle. We review the problem's history, reduce periodicity to a second brake, and establish nonperiodicity on an explicit real-parameter interval, a punctured near-isosceles neighborhood, and infinitely many thin-triangle windows. In Euclid's rational parameter u, validated flow enclosures and a terminal escape estimate prove nonperiodicity for every $u \in [0.3916, 0.4094]$, and hence for about 3.9% of all primitive triples ordered by hypotenuse, as well as for every $u \in [0.29, 0.2900000101]$. The Lagrange–Jacobi identity confines second brakes to strict maxima of the moment of inertia, where the remaining condition is vanishing shape velocity. Near the thin endpoint, matching to a restricted parabolic scattering problem and equidistribution of the binary phase give a set of positive lower density of nonperiodic members of $(4n^2 - 1, 4n, 4n^2 + 1)$. A uniform angular-momentum estimate also excludes first-turn brakes in the accumulating parabolic layer. Numerical continuation produces nearby periodic candidates with nonzero Pythagorean defects. The universal conjecture remains open.

There is no answer to the Pythagorean theorem. Well, there is an answer, but by the time you figure it out, I got 40 points, 10 rebounds and then we're planning for the parade.

— Shaquille O'Neal

1 A simple plan

Place point masses 3, 4, and 5 at the vertices of a right triangle with sides 3, 4, and 5, each mass opposite the side of the same length, and release them from rest under Newtonian gravity. The simple relation between mass and geometry suggests an orderly motion. The bodies undergo a succession of close encounters, repeatedly changing which pair is closest before one body escapes and the other two remain bound.

The possibility of periodic motion motivated the original experiment more than a century ago. Although numerical integration resolved the fate of the 3–4–5 triangle, the corresponding question for other integer right triangles remains open. Periodic free-fall orbits are known, and some lie close to the prescribed mass–side relation. The conjecture investigated here is that none satisfies that relation for an integer right triangle. The interest lies in this exact restriction: it links a familiar arithmetic family to a dynamical condition that depends on the entire subsequent motion.

1.1 From Meissel's question to computer experiments

In his 1913 paper, Carl Burrau recalled an 1893 conversation with the Kiel mathematician Ernst Meissel, who expected the 3–4–5 initial conditions to produce periodic motion. Burrau also used the problem to test variable-step numerical integration. His account credits Sigurd Kristensen with a substantial part of the calculation, which ended at $t = 3.35$, too early to determine the eventual outcome [1, 5].

Related work continued in the intervening decades. Strömgren studied periodic solutions for three comparable masses in 1919; Zumkley cited Burrau in a 1941 paper describing a slide-rule integration with three unit masses [3, 4]. Szebehely later discussed these studies and Garcia's 1966 extension of Zumkley's calculation. At Szebehely's suggestion, the original Pythagorean problem was taken up in 1966 by Standish at Yale, Spinelli with Lecar and Szebehely at NASA's Institute for Space Studies, and Stanek under Stiefel at ETH Zürich [2]. The sources reviewed here do not identify how Szebehely first encountered Burrau's work or establish an intervening extension of the exact 3–4–5 integration.

Szebehely and C. Frederick Peters followed the original motion far enough to identify its eventual behavior: the two heavier bodies, of masses 4 and 5, form a binary, while the mass-3 body escapes. Their 1967 numerical solution contradicted Meissel's expectation for this triangle [5]. In a separate paper that year, they found a nearby periodic solution with a binary collision and perturbed initial distances [6]. That example is outside the collision-free question considered here.

1.2 Periodic free fall and the Pythagorean restriction

Starting from rest does not itself rule out periodic motion. Standish's collisionless periodic example appeared in 1970 [7]. Kiyotaka Tanikawa, Hiroaki Umehara, and Hiroshi Abe developed systematic searches for binary- and triple-collision orbits in 1995; Tanikawa's 2000 continuation studied the structure of this free-fall initial-condition space [8, 9]. Moeckel, Montgomery, and Venturelli subsequently developed the rigorous passage from a configuration at rest to the first collinear configuration, and constructed periodic collision orbits in the isosceles problem [10]. Montgomery's *Dropping Bodies* gives a modern account of periodic free-fall motion and its reversal symmetry [12].

Li and Liao's 2019 catalogue reports 316 periodic free-fall solutions, including 313 new examples, for several mass ratios [13]. Hristov, Hristova, Dmitrašinović, and Tanikawa's 2024 equal-mass search reports 12,409 distinct solutions below its specified scale-invariant period cutoff [14]. These catalogues vary initial shapes more freely than our experiment. Equal masses cannot satisfy our right-triangle mass–side rule. Unequal-mass entries are useful starting guesses, but must still satisfy both the mass–side relation and the right-angle equation.

The original Pythagorean problem also remains a numerical benchmark: Rantala, Pihajoki, Mannerkoski, Johansson, and Naab use it to test their 2020 regularized integrator [15]. Chitan, Mylläri, and Haque revisited Pythagorean black-hole triples in 2022, and Chitan, Mylläri, and Valtonen studied spin effects in 2025 [16, 17]. Those relativistic extensions study encounters, escape, and mergers under equations different from the Newtonian system considered here.

1.3 One number specifies the experiment

The rule "mass equals opposite side" is understood in chosen mass and length units, with gravitational constant set to one. Enlarging all three masses and all three lengths by the same factor changes the clock rate, but preserves whether the motion repeats. We can therefore divide by the hypotenuse and work with a longest side and largest mass equal to one. Write the two other side lengths, and their corresponding masses, as A and B. The right-angle condition becomes $A^2 + B^2 = 1$.

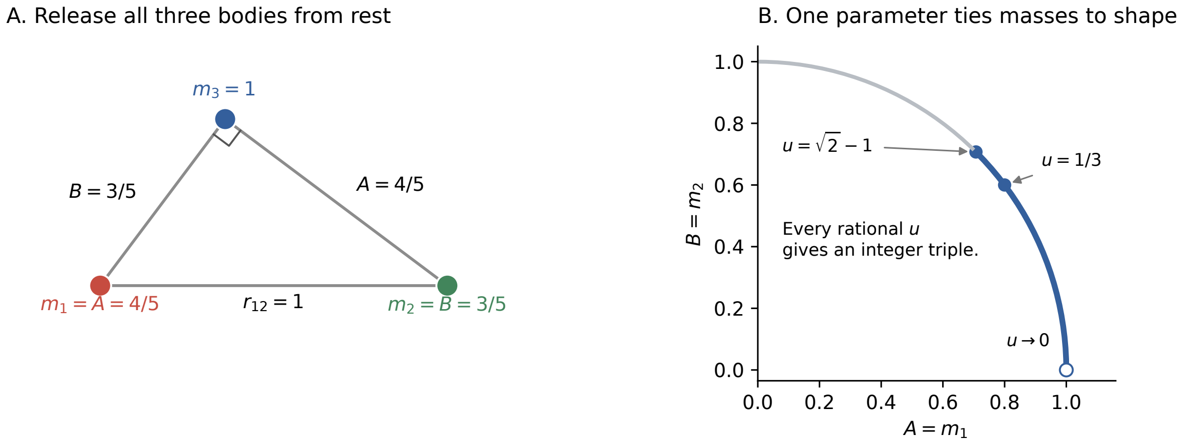

Figure 1: The experiment and its parameter. Left: the normalized $u = 1/3$ triangle, with masses $(4/5, 3/5, 1)$ opposite equal-numbered side lengths; all velocities initially vanish. Right: the allowed mass pairs lie on the quarter unit circle. The darker arc is the fundamental interval $0 < u \leq \sqrt{2} - 1$; the thin-triangle endpoint $u = 0$ is excluded. Rational u, rather than arbitrary points selected numerically on the arc, produces the integer triples.

The standard rational parameterization gives

$$A = \frac{1 - u^2}{1 + u^2}, \qquad B = \frac{2u}{1 + u^2}. \tag{1}$$

Rational u give precisely the integer right triangles up to common scale; Section 3 records the correspondence with Euclid's formula. For example, $u = 1/2$ gives $(3, 4, 5)$, while $u = 1/3$ gives $(4, 3, 5)$. Exchanging the legs only relabels the experiment, so we restrict to

$$0 < u \leq u_{\text{iso}} := \sqrt{2} - 1, \qquad A \geq B.$$

Near zero the triangle is very thin; at the other endpoint its two legs are equal. That equal-leg endpoint has irrational u, so it belongs to the continuous family used in the proof, but not to the integer conjecture. Changing u changes the masses and the initial shape together.

1.4 Brakes, periodicity, and current progress

Periodicity means that every labelled body returns to its initial position and velocity in a fixed inertial frame, without an intervening collision. Time reversal gives a useful characterization. If all three bodies stop simultaneously a second time, the trajectory retraces its path and returns to the initial state. Conversely, a periodic solution released from rest must have such a second stop, called a *brake*. Thus the conjecture amounts to excluding a second brake before the first collision. Section 2 describes our attempts to construct one.

A second brake can occur only at a strict local maximum of the moment of inertia, which measures the system's size about its center of mass. This reduces the search to particular events, but arbitrarily many close encounters may precede a possible return. Figure 2 illustrates the finite-time motion for nearby rational parameters and the progressively finer sampling needed to examine it.

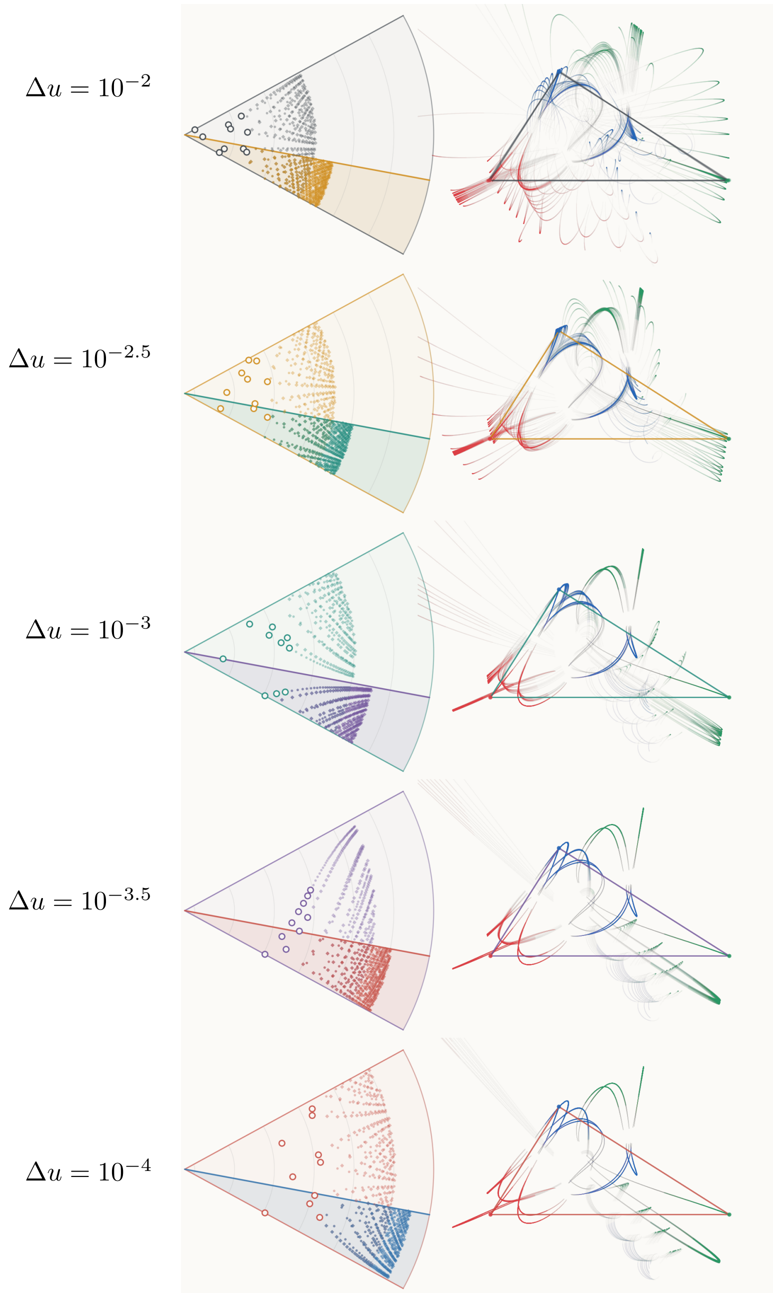

Figure 2: Progressively narrower collections of integer experiments near $u = 0.29$, using the project's original fan–trajectory visualization. From top to bottom the intervals are $[0.29, 0.29 + \Delta u]$, with $\Delta u = 10^{-2}, 10^{-2.5}, 10^{-3}, 10^{-3.5}, 10^{-4}$. Each left panel displays 400 primitive triples of smallest hypotenuse in its interval, with radius on a common affine $\log_{10} c$ scale and its angular interval expanded to one display radian. The highlighted sector is the next narrower interval; large rings select ten triples whose trajectories appear at right. Red, green, and blue track the three bodies in the source plot's leg ordering. Right panels share one spatial scale, show normalized physical times $0 \leq t \leq 4$, and clip outgoing paths. Opacity decreases with body speed; it is not an integration-error measure. Even the narrowest displayed interval is about 9,901 times wider than the certified interval. This ordinary numerical illustration conveys the search geometry, not validated parameter coverage.

Validated integration, followed by an escape estimate controlling all later times, excludes second brakes for every real $u \in [0.3916, 0.4094]$, an interval containing $(20, 21, 29)$ whose members and leg-exchanged mirrors make up about 3.9% of all primitive triples ordered by hypotenuse, and for every real $u \in [0.29, 0.2900000101]$. Point certificates settle $(7, 24, 25)$, $(9, 40, 41)$, and $(36, 77, 85)$. Time reversal about a second brake, combined with Montgomery's theorem on syzygies, shows that no second brake can occur before the first collinear instant and that any later one must sit inside a repeated syzygy letter (Section 5). Separate arguments prove nonperiodicity in a punctured neighborhood of the equal-leg endpoint and on infinitely many open intervals accumulating at the thin-triangle endpoint. The latter contain a positive-lower-density set of members of the primitive family $(4n^2 - 1, 4n, 4n^2 + 1)$, with density measured by the index n.

The near-equal-leg cutoff is still existential, and the thin-triangle argument leaves returning trajectories unclassified. Much of the remaining parameter interval is also open. We derive torque-history identities as a possible means of extending these results; their proposed sign argument requires a terminal estimate that has not been proved (Section 9). Neither a universal proof nor an exact counterexample is presently known.

1.5 Why measure zero does not settle the question

Let $\mathcal{P}$ be the set of real parameters in our fundamental interval whose collision-free trajectories are periodic. The integer conjecture is the exact assertion

$$\mathcal{P} \cap \mathbb{Q} = \varnothing,$$

not an assertion about the length, or Lebesgue measure, of $\mathcal{P}$. Even a proof that $\mathcal{P}$ has measure zero would leave its intersection with $\mathbb{Q}$ completely undetermined. Choosing a real number from a continuous probability distribution almost surely misses the rationals; that experiment does not model solving an exact dynamical equation.

The distinction persists within analytic, time-reversible Hamiltonian brake systems. Consider

$$H_u = \tfrac{1}{2}(p_x^2 + x^2 + p_y^2 + u^2 y^2), \qquad x(0) = y(0) = 1, \quad p_x(0) = p_y(0) = 0.$$

Here $x(t) = \cos t$ and $y(t) = \cos(ut)$. A second brake requires $t = n\pi$ and $ut = m\pi$, with positive integers m, n, and therefore

$$\text{a second brake exists} \quad \Longleftrightarrow \quad u = m/n \in \mathbb{Q}.$$

Almost every parameter is nonperiodic, yet *every rational parameter is periodic.* Nor would isolation suffice: the brake family $x(t) = \cos t$, $y(t) = (u - 1/3)\cos(\sqrt{2}\, t)$ has exactly one periodic parameter, the rational number $1/3$. These are not gravitational counterexamples. They show that analyticity, reversal symmetry, measure zero, and even isolated periodic parameters cannot by themselves supply the missing arithmetic obstruction.

There is a serious heuristic in favor of the conjecture: after removing translations and rotations, a simultaneous stop imposes three independent velocity conditions, while the prescribed family supplies only u and the stopping time. A transverse brake map would therefore have no zeros. But transversality for this particular gravitational family is precisely what is not known; a dimension count cannot assume it. Likewise, assigning a continuous random distribution to any isolated roots would exclude rational roots almost surely, but no such distribution is supplied by the dynamics.

The evidence for taking counterexamples seriously is more specific than this logical possibility. Section 2 gives six numerical second-brake candidates satisfying the mass–side rule. One has an initial triangle almost indistinguishable from 3–4–5, with its largest angle only 0.202° above a right angle. These are selected near misses, not random trials, and none meets the right-angle equation. They nevertheless show that periodic free fall and mass–side matching coexist numerically very close to the target. The unresolved long encounter sequences leave room for an exceptional exact match; neither the nearby orbits nor the measure argument assigns a probability to that possibility.

A subjective assessment (GPT-6 Astra, September 6, 2026). My best estimate is about a *10% chance* that at least one integer Pythagorean triangle produces an exactly periodic, collision-free motion. I favor the conjecture because an exact right angle and rational parameter must still accompany the second stop, with no known mechanism forcing them. I would nevertheless give a search a real chance: the nearby periodic candidates are substantial, and the longer encounter patterns are far from classified. This is a rough judgment about whether an example exists somewhere in the entire family, not a measured frequency or a prediction that a finite search will find one—roughly one chance in ten.

2 Attempts to construct a counterexample

The most suggestive near miss is a numerical periodic orbit whose launch triangle differs visibly little from 3–4–5 (Figure 3). Its masses equal its opposite initial sides to shooting accuracy, but its largest angle is about 90.202°. Understanding both this proximity and the remaining defect is essential: periodicity must survive imposing the exact right angle, and an integer counterexample must then have rational Euclid parameter.

2.1 Mass–side matching and the right-angle defect

Normalize $m_3 = r_{12} = 1$. In the larger family where each mass equals its opposite side, a valid nondegenerate triangle has

$$q_1 = (-1/2, 0), \quad q_2 = (1/2, 0), \quad q_3 = (x, y), \quad x = \frac{m_2^2 - m_1^2}{2}, \quad y = \sqrt{m_2^2 - (x + 1/2)^2} > 0. \tag{2}$$

The two mass ratios can vary independently. The remaining right-angle defect is

$$D = m_1^2 + m_2^2 - 1. \tag{3}$$

An exact zero of D is required even to challenge the stronger conjecture for real parameters. An integer counterexample additionally needs rational $u = m_2/(1 + m_1)$.

Our September 2026 campaign starts from six entries of Li and Liao's unequal-mass catalogue [13]. In the numerical construction we solve three independent second-brake equations together with two mass–side matching equations, varying the two initial-position coordinates, two mass ratios, and a regularized flight time. Levi–Civita (LC) coordinates replace the selected pair separation by a complex square and slow the clock near close approaches. An atlas of pair choices avoids relying on one coordinate chart for every encounter. Analytic variational equations supply the shooting derivatives; pseudo-arclength continuation recovers seeds for which direct correction stalls.

2.2 The near-3–4–5 orbit

In the catalogue's leg ordering, the exact normalized 3–4–5 data are $(m_1, m_2, m_3) = (0.6, 0.8, 1)$ and $q_3 = (0.14, 0.48)$, with the other vertices as in (2). Li and Liao's F_{30} entry has these exact

Seed	m_1	m_2	Second-brake time	D
F_1	.536940689	.839991651	1.337970014	$-.006108723$
F_2	.485763526	.878297063	1.413330015	$+.007371934$
F_3	.343754597	.887462426	1.074865097	$-.094243220$
F_4	.335421060	.903827720	1.133475174	$-.070588165$
F_5	.474742622	.896107795	1.544076262	$+.028389737$
F_{30}	.594811647	.801774973	6.289233824	$-.003355997$

Table 1: Ordinary numerical periodic candidates in the larger mass–opposite-side family. None is Pythagorean. Digits are displayed for reproducibility, not as interval-certified enclosures. The flight time is physical normalized time; twice this value would be a period at an exact collision-free second brake. Seed names refer to catalogue origins, not to established connected branches of solutions.

masses but $q_3 \simeq (0.1446319096, 0.4773197126)$ [13]. Its proximity is partly a selection effect: the catalogue already samples the 3–4–5 mass ratio, and this entry starts near the required triangle. Varying the two mass ratios to impose mass–side matching gives the continued candidate

$$\begin{aligned}(m_1, m_2) &\simeq (0.594811646571,\ 0.801774973318),\\ q_3 &\simeq (0.144521106471,\ 0.476902140016), \qquad \tau \simeq 6.289233824.\end{aligned} \tag{4}$$

Here τ is the second-brake time; an exact root would have period $2\tau \simeq 12.578467648$. The leg masses change from 0.6 and 0.8 by about -0.865% and $+0.222\%$. The angle γ opposite the unit side satisfies

$$\cos\gamma = \frac{D}{2m_1m_2}, \qquad D \simeq -0.003355997265, \qquad \gamma \simeq 90.2015966^\circ.$$

After an optimal labelled similarity fit, the initial triangle's RMS vertex distance from 3–4–5 is only 0.00223745, relative to the unit longest side. The figure makes that geometric proximity apparent.

The small defect is not a remaining free adjustment. The three brake conditions and two mass–side equations use all five shooting variables. The numerical Jacobian is nonsingular, suggesting local isolation rather than a tunable family; existence and isolation still require validation. Imposing $D = 0$ is a sixth equation. Projecting the displayed masses onto the unit circle changes the initial value problem and does not preserve its second brake. Moreover, finite decimal approximations establish neither rationality nor irrationality of an underlying exact root. A counterexample requires a mechanism that meets both constraints, not further digits of this near miss.

2.3 Numerical validation and its limits

The corrected shooting residuals are at most about 1.2×10^{-12}. Reintegration with an implicit Radau method, comparison with DOP853, and backward reversal reproduce the candidate states to roughly 2×10^{-12}. The corresponding five-equation Jacobians have ordinary smallest singular values between approximately 0.199 and 0.870 in the recorded physical-parameter convention. Their numerical nonsingularity suggests isolated solutions of the five matching equations. It does not prove existence or isolation: those conclusions require a validated implicit-function calculation.

The closest sampled separations range from about 3.4×10^{-6} to 5.0×10^{-4}. These values explain the use of regularization and also limit the evidence: sampling alone does not certify positive separation for every intervening time. Reconstructed energy is especially sensitive to cancellation near the deepest passages; the campaign's worst sampled physical energy error is about 2.5×10^{-7}. The shooting residuals consequently give no comparable error bound on the reconstructed energy.

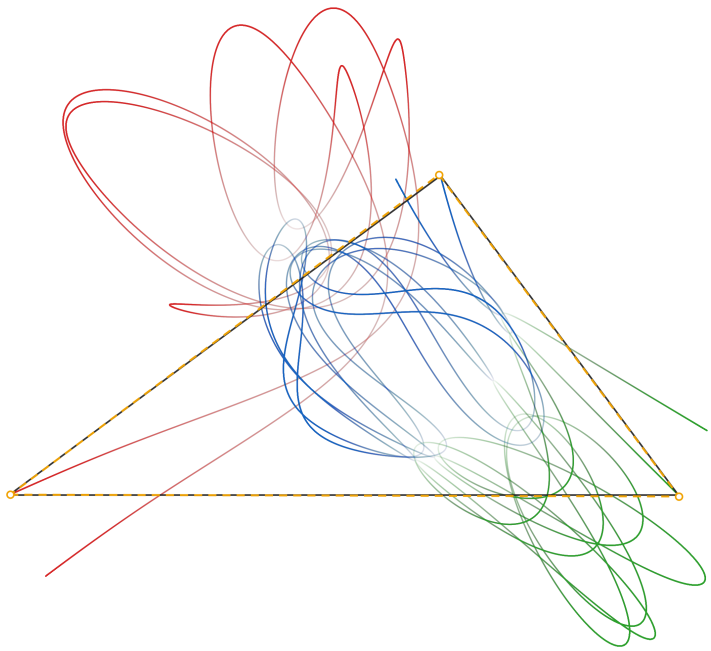

Figure 3: The continued F_{30} numerical periodic candidate near 3–4–5. Red, green, and blue trace masses $m_1, m_2, 1$ in the center-of-mass frame. The solid dark triangle is the launch configuration; the dashed gold triangle is the exact labelled 3–4–5 shape after least-squares translation, rotation, and common scaling (no reflection). Points are sampled uniformly in physical time. The plotted return half is constructed by reversing the first brake-to-brake segment; visual closure is not an independent periodicity test. This global-LC plotting replay has a maximum shooting residual of 2.6×10^{-11}; the campaign checks below use a separately refined root. Neither is an interval existence proof, and the nonzero right-angle defect rules out a counterexample.

We separately impose the exact right-triangle parameterization and look for small kinetic energy at local maxima of the moment of inertia. Four local searches seeded from F_1, F_2, F_3, F_5 converge to positive minima, the smallest about 1.28×10^{-4}. The F_{30} search reaches its evaluation limit and remains noise-limited; the F_4 event correction leaves the required strict-maximum branch. Neither incomplete search is an exclusion theorem. No exact right-triangle second brake was found, so there is no candidate on which rational reconstruction would be warranted. Machine-readable results and exact replay commands accompany the campaign report listed in Appendix A.

2.4 A systematic atlas of near misses

The global scan of Section 5 locates, for each stutter arc of the tied family, the local minimum over u of the minimal kinetic energy on that arc. Each such near miss is the shadow of a periodic brake orbit of the larger mass–side family nearby, and Newton continuation in (m_1, m_2, τ) on the complete Hopf residual recovers it. Seventeen distinct periodic orbits were obtained with residual below 10^{-6}, most below 10^{-9}; Table 2 lists the closest to the Pythagorean circle. Their defects are all nonzero. The smallest, $D \simeq -3.3 \times 10^{-6}$, corresponds to a largest angle of $90.0002°$; it is a thousand times closer to a right angle than F_{30}, and has a half period of 27.9 with 221 syzygies. The distribution of these minima, roughly one per stutter arc and decreasing in size only slowly with the number of arcs examined, is what the generic codimension-two heuristic predicts: the tied curve passes ever closer to periodic points as longer itineraries are included, without any indication of an exact crossing. These are ordinary numerical results (`data/mass_side_periodic_atlas.json`).

m_1	m_2	Half period	D	$\gamma - 90°$
.8027060912	.5963720697	27.8852	-3.29×10^{-6}	$-0.00020°$
.8773164381	.4798992402	4.7914	-1.26×10^{-5}	$-0.00086°$
.7740520242	.6331099219	7.7406	-1.53×10^{-5}	$-0.00089°$
.8814436238	.4722718402	7.1563	-1.65×10^{-5}	$-0.00113°$
.8419706850	.5394895948	6.6851	-3.63×10^{-5}	$-0.00229°$
.7946037376	.6070894201	9.0685	-4.73×10^{-5}	$-0.00281°$
.7946943401	.6070540791	10.7231	$+5.38 \times 10^{-5}$	$+0.00319°$
.8159034471	.5782736775	8.9566	$+9.89 \times 10^{-5}$	$+0.00600°$

Table 2: Periodic brake orbits of the mass–side family found by continuation from the tied family's near misses, ordered by right-angle defect. Masses are $(m_1, m_2, 1)$ with initial sides equal to the opposite masses; γ is the angle opposite the unit side. Double-precision shooting, residual below 10^{-7}; none is interval-validated and none is a counterexample.

2.5 Continuation, rank, and syzygy itineraries

The opposite defects of F_1 and F_2 would bracket a zero on a connected collision-free periodic branch joining them. No such branch is established. Their labelled half-orbit syzygy sequences differ. At zero angular momentum, every syzygy reached from a noncollinear initial state is transverse: a tangency would force both Jacobi velocities to lie on the collinear axis, and uniqueness would make the entire orbit collinear. Thus a finite syzygy itinerary is locally constant on a family of noncollinear brake endpoints with finite flight time and positive separation. Changing the itinerary requires leaving this domain.

Nor does a rank deficiency automatically construct the missing branch. For five smooth equations in five unknowns, a rank-four Jacobian leaves one direction to inspect, but the remaining scalar equation after Lyapunov–Schmidt reduction can be $s^2 = 0$, with an isolated root. A branch requires that residual equation to vanish along the proposed continuation. Similarly, raw free-group words from the catalogue do not distinguish these brake loops: a full brake orbit retraces its path and its loop in collision-free shape space is null-homotopic. The ordered, labelled syzygy sequence is the useful finite-itinerary information here.

The six corrected candidates all have nonzero Pythagorean defects. Their existence provides useful information about nearby periodic motion; the searches leave other itineraries and longer periods open.

3 The exact problem and its reductions

Let $a^2 + b^2 = c^2$ with positive integers, set $(m_1, m_2, m_3) = (a, b, c)$, and put

$$q_1 = (-c/2, 0), \quad q_2 = (c/2, 0), \quad q_3 = ((b^2 - a^2)/(2c), ab/c), \qquad \dot{q}_i = 0.$$

The side opposite each mass has the same length as that mass. We work with $G = 1$ and the equations

$$\ddot{q}_i = \sum_{j \neq i} m_j \frac{q_j - q_i}{|q_j - q_i|^3}.$$

Conjecture 1 (Pythagorean–Burrau nonperiodicity)**.** *For every positive integer triple $a^2+b^2=c^2$, the solution with these initial data has no collision-free labelled period.*

A labelled period is an exact return in the fixed inertial frame. The maximal classical solution ends at its first collision; regularized continuation through that collision is a different notion of solution.

3.1 Scaling and the rational parameter curve

Lemma 2 (Simultaneous mass–length scaling)**.** *If $q_i(t)$ solves the equations with masses m_i, then for $k>0$*

$$\widetilde{m}_i = km_i, \qquad \widetilde{q}_i(t) = kq_i(t/k)$$

solves the rescaled problem. A labelled period T corresponds exactly to a period kT.

Proof. Both sides of Newton's acceleration equation acquire the factor k^{-1}. Velocity is unchanged after the corresponding time substitution, and mutual distances acquire the positive factor k. $\square$

It follows that only primitive triples matter. For $u \in (0,1)$ define

$$A(u) = \frac{1-u^2}{1+u^2}, \qquad B(u) = \frac{2u}{1+u^2}.$$

If $u=p/q$ is reduced, Euclid's integer triple is $(q^2-p^2, 2pq, q^2+p^2)$ and its common divisor is two exactly when p,q are both odd. Conversely Euclid's theorem recovers every ordered primitive triple. The inverse is $u = B/(1+A)$, and leg exchange is the involution $u \mapsto (1-u)/(1+u)$. Its fixed point is $\sqrt{2}-1$, giving the fundamental interval $0<u\le\sqrt{2}-1$ for the real family.

After normalizing $c=1$, masses and positions are

$$(A,B,1), \qquad q_1=(-1/2,0), \quad q_2=(1/2,0), \quad q_3=((B^2-A^2)/2, AB).$$

We call this one-parameter family *tied*, since its masses and initial opposite sides vary together. The stronger real conjecture makes the same nonperiodicity assertion for every $0<u\le\sqrt{2}-1$.

3.2 Initial geometry

The initial center of mass is the incenter: both have barycentric weights equal to the opposite side lengths. The identity $I = (\sum m_i)^{-1}\sum_{i<j} m_i m_j r_{ij}^2$ gives, directly,

$$I_0 = abc, \qquad U_0 = \frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a}.$$

In the normalized problem,

$$I_0 = AB, \qquad U_0 = AB + (AB)^{-1}, \qquad H = -U_0.$$

Here I is the moment of inertia about the center of mass and U is the positive potential magnitude. The right angle is not preserved: for $D = r_{12}^2 - r_{23}^2 - r_{31}^2$,

$$D(0) = \dot{D}(0) = 0, \qquad \ddot{D}(0) = 2\big[(AB)^{-1} - AB(A+B)\big] > 0.$$

Thus the angle opposite r_{12} initially becomes obtuse. This local sign does not define an invariant region; numerical integration at $u=1/3$ finds later changes of sign of D.

3.3 Second brakes and a residual valid at collinearity

Lemma 3 (Second-brake lemma)**.** *A classical free-fall segment generates a collision-free labelled periodic solution if and only if all inertial velocities vanish at some positive time before collision.*

Proof. Uniqueness and time reversal give $q(-t) = q(t)$. A brake at τ gives $q(\tau + s) = q(\tau - s)$ and hence a labelled period 2τ. Conversely, a labelled period T combined with the symmetry at zero gives reflection symmetry about $T/2$, and therefore zero velocity there. The period 2τ need not be minimal unless τ is the first positive brake. $\square$

Write $r_{ij} = |q_j - q_i|$, $K = \frac{1}{2}\sum_i m_i|\dot{q}_i|^2$, $U = \sum_{i<j} m_i m_j/r_{ij}$, and $H = K - U = -U_0$. In the center-of-mass frame define reduced masses $\mu_1 = m_1m_2/(m_1+m_2)$ and $\mu_2 = m_3(m_1+m_2)/(m_1+m_2+m_3)$, and Jacobi vectors

$$X = q_2 - q_1, \qquad Y = q_3 - \frac{m_1q_1 + m_2q_2}{m_1 + m_2}$$

and Hopf invariants

$$h(X,Y) = \left(\frac{|X|^2 - |Y|^2}{2}, X \cdot Y, X \times Y\right).$$

We use the quotient residual

$$\mathcal{B}(u,t) = \frac{d}{dt}h(X(u,t), Y(u,t)).$$

Away from triple collision, Dh has rank three and kernel equal to the common rotation direction. Since total linear and angular momentum vanish, $\mathcal{B} = 0$ is equivalent to vanishing of every labelled inertial velocity. Unlike three mutual-distance derivatives, this remains true at syzygy. Indeed, a velocity in $\ker Dh$ has the form $(\dot{X}, \dot{Y}) = \omega(iX, iY)$, whose angular momentum is ωI; $L = 0$ and $I > 0$ force $\omega = 0$.

For an explicit configuration-space reduction, rotate with X and write $X = (R, 0)$, $Y = (C, D)$. Eliminating the common rotation at $L = 0$ gives

$$K_{\mathrm{red}} = \frac{\mu_1\dot{R}^2 + \mu_2(\dot{C}^2 + \dot{D}^2)}{2} - \frac{\mu_2^2(C\dot{D} - D\dot{C})^2}{2[\mu_1R^2 + \mu_2(C^2 + D^2)]}.$$

Here the dots on C, D differentiate coordinates in the moving frame. Substitution of $\dot{\theta} = -\mu_2(C\dot{D} - D\dot{C})/I$ into the inertial kinetic energy proves the formula. With $m_{12} = m_1 + m_2$, the potential is

$$U = \frac{m_1m_2}{R} + \frac{m_1m_3}{\sqrt{(C + m_2R/m_{12})^2 + D^2}} + \frac{m_2m_3}{\sqrt{(C - m_1R/m_{12})^2 + D^2}}.$$

The tied brake curve is $R = 1$, $C = AB(B - A)/(A + B)$, $D = AB$, with zero reduced velocity.

4 Finite certificates for nonperiodicity

Let

$$I = \mu_1|X|^2 + \mu_2|Y|^2, \qquad \sigma = \mu_1\overline{X}\dot{X} + \mu_2\overline{Y}\dot{Y}, \qquad \zeta = \mu_1\overline{X}\dot{X} - \mu_2\overline{Y}\dot{Y}.$$

Here complex multiplication identifies the oriented plane with $\mathbb{C}$. Then $\dot{I} = 2\,\mathrm{Re}\,\sigma$ and the angular momentum is $L = \mathrm{Im}\,\sigma$. Consequently, at $L = 0$ and away from $Y = 0$,

$$\dot{I} = 0 \ \text{ and } \ \zeta = 0 \quad \Longleftrightarrow \quad \dot{X} = \dot{Y} = 0.$$

The implication from a brake to these three scalar equalities is unconditional. At the Jacobi degeneracy $Y = 0$, the Hopf residual $\mathcal{B}$ above supplies the coordinate-independent converse.

This splitting has an exact shape interpretation. Put $J_X = \mu_1|X|^2$, $J_Y = \mu_2|Y|^2$, $s = J_X/I$, and $\phi = \arg Y - \arg X$. At every zero of $\dot{I}$ with $X, Y \neq 0$ and $L = 0$,

$$\boxed{\zeta = I\left(\dot{s} - 2is(1-s)\dot{\phi}\right), \qquad |\zeta|^2 = \frac{8KJ_XJ_Y}{I} = 8KI\,s(1-s).}$$

Indeed $\sigma = 0$ gives $\zeta = 2\mu_1\overline{X}\dot{X} = -2\mu_2\overline{Y}\dot{Y}$; decomposing each Jacobi velocity into radial and angular parts proves both formulas. Thus a maximum event is a brake exactly when its shape velocity vanishes, and the origin-avoidance problem can equivalently be written as the scalar strict inequality $K = U - U_0 > 0$. At a brake time τ, time reversal gives

$$K(\tau + s) = \frac{s^2}{2}\sum_i m_i|\ddot{q}_i(\tau)|^2 + O(s^4).$$

The coefficient is positive, since $\ddot{I}(\tau) = -2U_0 < 0$ rules out vanishing of all accelerations. A brake is consequently a quadratic zero of K in time, which cannot be detected by a sign change of K.

There is also a brake residual intrinsic to any binary Levi–Civita chart. For a selected pair write $g = q_j - q_i = w^2$, $dt = |w|^2 d\sigma$, and $z = dw/d\sigma$. Let G be the third body relative to the selected pair's center of mass and $P = \dot{G}$. With reduced masses μ_g, μ_G, angular momentum is

$$L = 2\mu_g \operatorname{Im}(\overline{w}z) + \mu_G G \times P.$$

Thus, on a collision-free zero-angular-momentum trajectory and wherever $G \neq 0$,

$$\boxed{(z_r, z_i, G \cdot P) = 0 \quad \Longleftrightarrow \quad \text{a labelled brake.}}$$

Indeed $z = 0$ gives $\dot{g} = 0$; then $L = 0$ gives $G \times P = 0$, while $G \cdot P = 0$ forces $P = 0$. Total momentum removes the common translation. This residual stays regular through arbitrarily close selected-pair passages; at $G = 0$ one changes Jacobi tree or uses $\mathcal{B}$.

Theorem 4 (Brake-event identities)**.** *Along a collision-free free-fall solution with initial potential U_0, every brake satisfies*

$$K = 0, \qquad U = U_0, \qquad \ddot{I} = -2U_0 < 0, \qquad r_{ij} \geq \frac{m_i m_j}{U_0}.$$

At any zero of $\dot{I}$ for which $\ddot{I} \geq 0$, one instead has $K \geq U_0$.

Proof. With positive potential magnitude U and energy $H = K - U = -U_0$, the Lagrange–Jacobi identity is

$$\ddot{I} = 4K - 2U = 2K - 2U_0 = 2U - 4U_0.$$

At a brake $K = 0$, giving the first three assertions. Since every summand $m_i m_j / r_{ij}$ is positive and bounded above by $U = U_0$, the separation bound follows. If $\ddot{I} \geq 0$, the middle expression instead gives $K \geq U_0$. □

Corollary 5 (Validated-cover criterion)**.** *Suppose a classical free-fall solution has a launch window $[0, t_L]$ on which $U < 2U_0$. Suppose validated flow enclosures cover every time in $[t_L, t_*]$, and every enclosure proves at least one of*

$$\dot{I} \neq 0, \qquad K > 0, \qquad \mathcal{B} \neq 0.$$

Then there is no second brake in $(0, t_*]$. If the terminal enclosure satisfies a proved forward collision-or-escape criterion that excludes every later brake, the initial free-fall solution is nonperiodic.

Proof. At launch $\dot{I}(0) = 0$, whereas $U < 2U_0$ implies $\ddot{I} < 0$. Consequently $\dot{I}(t) < 0$ for $0 < t \le t_L$. The remaining disjunction excludes a brake at every covered time. This argument applies fiber by fiber when t_* depends on a parameter; both section images and the complete intervening flow tubes must be covered. The initial brake itself is never subjected to the nonvanishing disjunction. ☐

This criterion has been implemented with exact rational initial data and MPFR interval arithmetic. The archived certificates prove the conjecture for

$$(21, 20, 29),\ (5311, 5280, 7489),\ (8319, 8200, 11681),\ (33439, 32400, 46561),\ (72, 65, 97)$$

in both leg orderings. The general multi-chart Levi–Civita certificate of Theorem 10 below, run on a single exact rational, further proves the conjecture for

$$(7, 24, 25),\ (9, 40, 41),\ (36, 77, 85),$$

whose light body is ejected after a single regularized passage of the heavy pair at separations down to 5×10^{-9}, and, in its multiprecision form (330 and 200 bits), for the classical 3–4–5 problem itself, by a fresh single-chart certificate through the 8×10^{-5} passage to the escape at $t \simeq 12.08$, for $(12, 35, 37)$, through a temporary $\{2, 3\}$ binary of seventy revolutions and an exchange, and for $(39, 80, 89)$, through seven chart legs and a final triple close approach in which all three separations fall below 0.025 (`docs/SMALL_TRIPLE_CERTIFICATES.md`; every log is replayed at a second Taylor order, tolerance, and, where multiprecision, working precision). The parameter-cover results below require additional control of how the enclosed trajectories depend on u.

Lemma 6 (Terminal escape certificate)**.** *For a prospective binary of total mass M and an outer body of mass m_c, let x, y be the inner and outer Jacobi vectors, $r = |x|$, $\rho = |y|$, and $e = |\dot{x}|^2/2 - M/r$. Choose $\eta > 0$, set $R = M/\eta$ and $s_0 = \rho_0 - R$, and suppose*

$$s_0 > 0, \quad \dot{\rho}_0 > 0, \quad \delta = \frac{1}{2}\dot{\rho}_0^2 - \frac{M + m_c}{s_0} > 0,$$

$$e_0 + \frac{m_c\sqrt{2MR}}{\sqrt{2\delta}\, s_0^2} < -\eta.$$

Then the future solution either ends in inner binary collision or the outer body escapes with $\dot{\rho} \ge \sqrt{2\delta}$ forever. In particular there is no later classical brake.

Proof. On the bootstrap region $e < -\eta$, one has $r < R$ and $r|\dot{x}| < \sqrt{2MR}$. The Newton field derivative bound $\|D(v/|v|^3)\| \le 2/|v|^3$ gives

$$|\dot{e}| \le \frac{2m_c r|\dot{x}|}{(\rho - r)^3}.$$

The outer Jacobi acceleration obeys $|\ddot{y}| \le (M+m_c)/(\rho-r)^2$, so $\ddot{\rho} \ge -(M+m_c)/(\rho-R)^2$. Therefore

$$J = \frac{1}{2}\dot{\rho}^2 - \frac{M + m_c}{\rho - R}$$

is nondecreasing while $\dot{\rho} > 0$. Its initial value $\delta > 0$ prevents $\dot{\rho}$ from reaching zero. Hence $\rho \ge \rho_0 + \sqrt{2\delta}(t - t_0)$, and the total future tidal increase of e is at most the allowance in the hypothesis. Its strict margin closes the bootstrap. The positive gap $\rho - r$ excludes an outer collision; only an inner binary collision can terminate the resulting escape trajectory. ☐

The near-triple-collision scaling below has limiting inner energy zero, so it requires an escape criterion that permits an expanding inner pair.

Lemma 7 (Escape with a possibly expanding inner pair). *Use the same Jacobi variables, let* $w_0 = \rho_0 - r_0 > 0$*, and choose* $\varepsilon, c > 0$*. Put*

$$v_b = \sqrt{2M/r_0 + 2\varepsilon}, \qquad \mathcal{T} = \int_0^\infty \frac{\sqrt{2M(r_0 + v_b t) + 2\varepsilon(r_0 + v_b t)^2}}{(w_0 + ct)^3}\, dt.$$

If

$$\dot\rho_0 - \frac{M + m_c}{c w_0} > v_b + c, \qquad e_0 + 2m_c\mathcal{T} < \varepsilon,$$

then, until a possible inner collision, $r(t) \le r_0 + v_b t$*,* $\dot\rho(t) > v_b + c$*, and* $\rho(t) - r(t) \ge w_0 + ct$*. In particular there is no later brake.*

Proof. Assume $e < \varepsilon$ and $\rho - r \ge w_0 + ct$ on a bootstrap interval. At a first outward crossing of $r_0 + v_b t$, the energy inequality gives $\dot r \le |\dot x| < v_b$, excluding that crossing. Integration of $\ddot\rho \ge -(M + m_c)/(w_0 + ct)^2$ then gives $\dot\rho > v_b + c$ and preserves the gap. Finally $r|\dot x| \le \sqrt{2Mr + 2\varepsilon r^2}$ bounds the total tidal increase of e by $2m_c\mathcal{T}$. The integral converges, and the strict energy margin closes the bootstrap. □

Theorem 8 (Validated middle first-maximum interval). *For every real*

$$\frac{29}{100} \le u \le \frac{14501}{50000},$$

the tied classical trajectory is collision-free through its first positive local maximum of I*, and this maximum is not a labelled brake. The unique intervening minimum and the first subsequent maximum obey*

$$t_{\min} \in [0.758038, 0.758045], \qquad t_{\max} \in [1.32908, 1.33943].$$

Proof. The exact tied launch is embedded as a one-dimensional mean-value graph with u retained as a frozen state coordinate. A pinned MPFR/CAPD C^1 Poincare calculation propagates this graph first in a pair–13 Levi–Civita chart and, after an exact Jacobi-tree transformation at $t = 1$, in a pair–23 chart. On every accepted-step enclosure, all three mutual separations are strictly positive. The launch-concavity window and whole-leg audits give $\dot I < 0$ after launch to the first minus-to-plus zero and $\dot I > 0$ from that zero to the first later plus-to-minus zero. At the minimum, $U/U_0 \in [5.17942, 5.18335]$; at the maximum, $U/U_0 \in [0.983016, 1.01907] \subset (0, 2)$, proving both events strict by Lagrange–Jacobi. In the final selected-pair chart the validated residual has

$$\operatorname{Im} z \in [-0.0500738, -0.0113187], \qquad P_x \in [-0.0561842, -0.00917817].$$

A labelled brake forces $z = P = 0$, so either strict component excludes it. Two abutting tiles cover the stated interval. The exact chart identities, logs, and an independent left-tile replay are indexed in Appendix A. □

This theorem excludes only the first positive maximum. It does not rule out a brake on a later maximum branch and therefore is not, by itself, a nonperiodicity theorem on the displayed interval.

Theorem 9 (Validated middle real interval)**.** *For every real*

$$\frac{29}{100} \le u \le \frac{2900000101}{10000000000},$$

the tied maximal classical solution has no second labelled brake at any collision-free time and hence is not a classical periodic solution. Consequently the Pythagorean–Burrau conjecture holds for infinitely many primitive triples in this interval, including $(9159, 5800, 10841)$ *at* $u = 29/100$*.*

More explicitly, for every integer $k \ge 99010$ set

$$u_k = \frac{290k+1}{1000k} = \frac{29}{100} + \frac{1}{1000k}.$$

The numerator and denominator are coprime and of opposite parity, and the interval inequality is equivalent to $1/(1000k) \le 101/10^{10}$. The theorem therefore proves nonperiodicity for the primitive triples

$$\left(915900k^2 - 580k - 1, 580000k^2 + 2000k, 1084100k^2 + 580k + 1\right), \qquad k \ge 99010.$$

Proof. A pinned MPFR/CAPD propagation embeds the exact launch curve on $[0.29, 0.290000001]$ in one correlated tripleton with u frozen as a state coordinate. Before the exchange, every accepted full-step enclosure is checked for three positive mutual separations. The launch window verifies $U < 2U_0$; subsequent windows verify the brake-exclusion disjunction of Corollary 5. Three algebraic changes of Levi–Civita chart use rigorous mean-value images.

Across the exchange, six C^1 Poincare maps project onto the transverse pair–13 sections

$$w_r = -\frac{3}{5}, -\frac{1}{2}, -\frac{2}{5}, -\frac{3}{10}, -\frac{1}{5}, 0.$$

For each map a convex mean-value domain contains the propagated family and its stored center, the validated section derivative transports the distinguished parameter generator, and an independent C^0 integration covers the complete incoming tube through the latest possible return. Thus section reconditioning does not omit any inter-section time. Every tube enclosure retains positive separation and excludes a brake.

For the terminal binary $\{2,3\}$ set $M = B+1$, $G = q_1 - C_{23}$, $\rho = |G|$, $h = |\dot q_3 - \dot q_2|^2/2 - M/r_{23}$, and

$$R = M/4, \quad d = \rho - R, \quad E_\rho = \frac{\dot\rho^2}{2} - \frac{A+B+1}{d}, \quad \Delta = \frac{A\sqrt{2MR}}{\sqrt{2E_\rho}\, d^2}.$$

These are precisely the quantities of Lemma 6; h is transported as a regularized state variable. Hence the terminal check can use its enclosure without dividing by the small inner separation. The resulting terminal enclosure has

$$\begin{aligned} &t_* \in [4.30030837364, 4.30032429905],\\ &d > 1.8655, \qquad \dot\rho > 2.0484, \qquad E_\rho > 0.8223,\\ &h < -9.2745, \qquad -4 - h - \Delta > 5.0690. \end{aligned}$$

These are the strict hypotheses of Lemma 6 with binary $\{2,3\}$, escaper body 1, and $\eta = 4$. Hence after t_* the inner pair either collides classically, ending the maximal classical solution, or body 1 escapes permanently; neither alternative permits a later brake. The second-brake lemma proves nonperiodicity. A separate interval checker verifies the terminal inequalities on an outward-widened box; an independent MPFR replay reproduces the section audits and escape margin.

The remaining extension is the union of 100 independently certified, adjacent closed tiles of exact width 10^{-10}, beginning at 0.2900000001 and ending at 0.2900000101. Every requested interval is identical to its PASS interval, every consecutive pair shares its exact rational endpoint, and all 100 full logs are archived. The campaign overlaps the initial tile, and their union is exactly the stated closed interval. $\square$

This continuum theorem covers a width of 1.01×10^{-8}.

Theorem 10 (Validated prompt-ejection window)**.** *For every real*

$$\frac{3916}{10000} \leq u \leq \frac{4094}{10000},$$

and hence, after leg exchange, for every real $u \in [0.419044\ldots, 0.437195\ldots]$*, the tied maximal classical solution has no second labelled brake at any collision-free time and is not a classical periodic solution.*

Corollary 11 (A positive proportion of all triples)**.** *The Pythagorean–Burrau conjecture holds for every primitive triple whose Euclid parameter* $b/(a+c)$*, in either leg ordering, lies in the two intervals of Theorem 10; these form a set of asymptotic density about* 3.9% *among all primitive Pythagorean triples ordered by hypotenuse, and contain* $(20, 21, 29)$.

Proof of the corollary. Primitive triples with hypotenuse at most N correspond to coprime opposite-parity lattice points (p, q) with $q > p > 0$ and $p^2 + q^2 \leq N$, and $u = p/q$. Lattice points with these multiplicative constraints are asymptotically equidistributed in angle within the quarter disc, so the density of the two intervals is their angular measure $(\arctan 0.4094 - \arctan 0.3916) + (\arctan 0.437195 - \arctan 0.419044) \simeq 0.0307$ divided by $\pi/4$, about 3.906%; a direct count gives 12,433 of the 318,320 primitive triples with hypotenuse at most 2×10^6. $\square$

Proof of the theorem. On this window the tied motion is a triple close approach in which pair $\{1,3\}$ passes at separation 0.02–0.05 near $t \simeq 0.47$, pair $\{2,3\}$ passes at 0.017–0.05 near $t \simeq 0.53$, and body 2 is then ejected leaving $\{1,3\}$ bound. The certificate integrates each closed rational tile as one correlated tripleton with u frozen as an exact affine state coordinate, first in the pair-$\{1,3\}$ Levi–Civita chart of Section 4, then, after an exact algebraic chart switch at physical time 0.5025 (or 0.505 above $u = 0.406$), in the pair-$\{2,3\}$ chart through its passage, and after a second switch at 0.575 again in the pair-$\{1,3\}$ chart. Regularized time synchronizes each passage across the tile; a fixed-clock representation fails through the same passages for tiles wider than about 10^{-8}.

Every accepted step verifies positive separations on its enclosure and, after the launch window on which $U < 2U_0$ holds, the brake-exclusion disjunction of Corollary 5 in the form $K > 0$, $\dot{I} \neq 0$, or $(z, P) \neq 0$ in the current chart. The run ends at the first regularized step at which the phase-robust form of Lemma 6, with binary $\{1,3\}$, escaper body 2, transported inner energy h, and the outer variables only, holds on the entire hull with rigorous margin above 0.05 for some $\eta \in \{1, \ldots, 8\}$; this is simultaneous for all fibers and excludes every later classical brake. Tiles have width 10^{-6} on $[0.3916, 0.4085]$ and 10^{-7} on $[0.4085, 0.4094]$, with a failing tile bisected; consecutive tiles share exact rational endpoints. All 25,900 tiles passed with minimum margin 0.0500; the logs, the campaign driver, and an independent log auditor that also replays a random sample of tiles at a different Taylor order and tolerance are indexed in Appendix A (`docs/ISO_WINDOW_INTERVAL.md`). $\square$

This window is about 1.8×10^6 times wider than the middle interval above. Its boundaries are dynamical: below $u \simeq 0.3914$ body 2 is no longer promptly unbound, and above $u \simeq 0.40945$ a chaotic sliver begins. Larger parameter covers remain a separate validation problem; experimental wider propagations are not included in its conclusion.

At a possible brake, the bound $r_{ij} \geq m_i m_j / U_0$ is uniform on every compact parameter interval bounded away from $u = 0$. It gives no lower bound on separations between brake events and degenerates as $u \to 0$.

On any regular strict-maximum branch, the implicit-function theorem locally gives $t = t_k(u)$. The unresolved condition becomes the planar origin-avoidance problem

$$u \longmapsto \zeta(u, t_k(u)) \in \mathbb{C} \setminus \{0\}.$$

To turn this into a universal theorem one must also control branch creation, collision and escape endpoints, degenerate events, and the possible number of branches. The nominal dimension count alone supplies none of these facts.

Numerical diagnostics at $u = 1/3$ place maximum-event values of ζ in all four quadrants, with the origin in the convex hull of three early values. These data argue against a fixed real linear projection with one strict sign at every maximum. Branch-dependent order or a nonlinear constraint remains a possible approach.

5 Syzygies, reversal, and where a second brake can occur

Time reversal about a second brake, combined with Montgomery's theorems on syzygies, restricts where a second brake can lie. A syzygy is a collinear instant. Every syzygy of a zero-angular-momentum orbit launched from a noncollinear rest configuration is transverse, and every brake of such an orbit is noncollinear (Section 2); so syzygies are isolated.

Theorem 12 (Montgomery [11])**.** *Every bounded, collision-free, zero-angular-momentum solution of the planar three-body problem, with any positive masses, has infinitely many syzygies. Moreover the normalized signed area*

$$z = \frac{4\sqrt{3}\,\Delta}{r_{12}^2 + r_{23}^2 + r_{31}^2}, \qquad \Delta = \tfrac{1}{2}(q_2 - q_1) \times (q_3 - q_1),$$

satisfies $\frac{d}{dt}(f\dot{z}) = -qz$ *along every zero-angular-momentum solution away from triple collision, with* $f > 0$ *and* $q \geq 0$*, where* q *vanishes only on the Lagrange homothetic solutions.*

Suppose the tied orbit brakes again at $\tau > 0$ after a collision-free segment. Uniqueness and reversal give $q(\tau + s) = q(\tau - s)$ and $q(2\tau + s) = q(s)$: the orbit is 2τ-periodic, bounded, and collision-free for all time. Let $0 < s_1 < \cdots < s_n < \tau$ be its syzygies before τ. Reflection sends them to the syzygies $2\tau - s_k$ of $(\tau, 2\tau)$, with $q(2\tau - s_k) = q(s_k)$ and $\dot{q}(2\tau - s_k) = -\dot{q}(s_k)$.

Theorem 13 (No brake before the first syzygy)**.** *For every real* u*, and for every noncollinear brake initial condition with any positive masses, there is no second brake at or before the first positive syzygy time.*

Proof. A brake is noncollinear, so a brake at or before the first syzygy occurs strictly before it, with $n = 0$. By reflection the periodic orbit has no syzygy in $(\tau, 2\tau)$ either, nor at 0, τ, 2τ. It is bounded, collision-free, and has zero angular momentum, yet has no syzygy at all, contradicting Montgomery's theorem. □

This settles unconditionally the first-arc objective of the torque-history route of Section 9; the corrected conditional lemma and its terminal hypothesis are not needed for that purpose.

Theorem 14 (A second brake is a stutter)**.** *If the orbit brakes at τ with $n \geq 1$ earlier syzygies, the two syzygies adjacent to τ, at s_n and $2\tau - s_n$, have the same collinear configuration and opposite velocities; in particular the same middle body. The syzygy sequence of the period is the double palindrome $(\sigma_1 \cdots \sigma_n)(\sigma_n \cdots \sigma_1)$ repeated, so the forward sequence contains a repeated adjacent letter (a stutter, in the terminology of [10]) at every brake.*

Hence an inter-syzygy arc whose two syzygies have different middle bodies contains no brake, without any further estimate. This is the converse of the observation in [10] that brake orbits run backwards produce an open set of stuttering orbits.

Theorem 15 (Position within the arc)**.** *Between consecutive syzygies z has exactly one critical point, a strict extremum. A second brake lies at that extremum of its stutter arc, and I has a strict maximum there.*

Proof. Where $z > 0$, $f\dot{z}$ is nonincreasing, so $\dot{z}$ changes sign at most once, and $\dot{z} \equiv 0$ on an interval would force $qz = 0$, that is a Lagrange homothetic motion. All velocities vanish at a brake, so $\dot{z}(\tau) = 0$; the inertia statement is Lagrange–Jacobi. $\square$

Theorem 16 (Brakes as fixed points on the syzygy section)**.** *Let Σ be the transverse syzygy states, P the next-syzygy map on its domain in Σ, and $R(q, v) = (q, -v)$. Then $RPR = P^{-1}$, so RP is an involution, and the inter-syzygy arc starting at $\sigma \in \Sigma$ contains a brake if and only if $RP(\sigma) = \sigma$.*

Proof. Reversibility gives $RPR = P^{-1}$. If $P(\sigma) = R\sigma$ on an arc of duration T, then $q(T) = q(0)$ and $v(T) = -v(0)$; uniqueness applied to $t \mapsto q(T - t)$ gives $q(T/2 + s) = q(T/2 - s)$, hence $v(T/2) = 0$. The converse is Theorem 14. $\square$

The fixed set of RP is exactly the set of syzygy states of brake-to-syzygy arcs run backwards, that is, the image of the syzygy map of [10] together with its velocities. For the tied family the second-brake question becomes an intersection problem between the one-parameter curves $u \mapsto \sigma_k(u)$ of kth syzygy states and this two-dimensional fixed set inside the four-dimensional reduced section. The codimension count is unchanged, but the brake-reachable states are now an explicit geometric object, and Conjecture 1 of [10] (that $\dot{I} < 0$ from a brake through its first syzygy) would add that I is strictly monotone on each half of a brake's stutter arc.

These statements are consistent with the global numerical picture of the tied family. In a scan of about 170,000 parameters on $[0.10, 0.4142]$ with steps down to 10^{-6}, every inter-syzygy arc had exactly one z-extremum, the first inertia minimum never preceded the first syzygy, about half of the escaping trajectories had no stutter before the escape lemma of Section 4 applied, and all of the forty closest approaches of the family to a second brake occurred on stutter arcs, as Theorem 14 predicts by continuity. The closest approaches found, after refinement to about 10^{-13} in u, are at $u \simeq 0.2003099995$, $t \simeq 7.91$, and $u \simeq 0.2556321826$, $t \simeq 4.79$, where the kinetic energy at its local minimum is only $7 \times 10^{-9}\, U_0$ and $8 \times 10^{-9}\, U_0$ respectively (residual velocities of order 2×10^{-4}); neither is a zero, and neither lies near a rational of small denominator beyond chance level. Section 2 uses these near misses as seeds.

6 The equal-leg endpoint and its neighbors

Theorem 17. *At $u = \sqrt{2} - 1$, the classical solution has no positive second brake and reaches a binary or triple collision in finite time.*

Proof. Here $m_1 = m_2 = m = 1/\sqrt{2}$ and reflection symmetry persists. Write the base bodies as $(-x, y_b), (x, y_b)$ and the apex as $(0, y_a)$, with $h = y_a - y_b$. Then

$$\ddot{x} = -\frac{m}{4x^2} - \frac{x}{(x^2 + h^2)^{3/2}} < 0.$$

Starting with $x_0 = 1/2$ and $\dot{x}(0) = 0$ gives $\dot{x} < 0$, excluding a second brake. While $x > 0$, one has $\ddot{x} \le -m/(4x_0^2)$, whence

$$x(t) \le x_0 - \frac{m}{8x_0^2} t^2.$$

Thus collision occurs no later than $\sqrt{8x_0^3/m} = 2^{1/4}$. $\square$

To study nearby rational parameters, we use the Levi–Civita continuation of the symmetric orbit as a comparison solution. A collision of a nearby physical trajectory still terminates its classical motion.

Put $v = u/(\sqrt{2} - 1)$, $(m, n) = (A, B)$, $g = q_2 - q_1 = w^2$, and $G = q_3 - C_{12}$. With $dt = |w|^2 d\sigma$, write $z = w_\sigma$, $P = \dot{G}$, and $e = |\dot{g}|^2/2 - (m + n)/|g|$. The selected pair collision is $w = 0$; this is regular in the comparison equations, but it still terminates the classical trajectory at $v = 1$.

For $d_{31} = G + ng/(m + n)$, $d_{23} = G - mg/(m + n)$, and $F = d_{23}/|d_{23}|^3 - d_{31}/|d_{31}|^3$, the regular equations are

$$\begin{aligned} w_\sigma &= z, \qquad z_\sigma = \frac{e}{2} w + \frac{|w|^2 \bar{w}}{2} F, \qquad e_\sigma = 2\,\mathrm{Re}(wz\bar{F}), \\ G_\sigma &= |w|^2 P, \quad P_\sigma = -|w|^2 \frac{m + n + 1}{m + n} \left(\frac{m d_{31}}{|d_{31}|^3} + \frac{n d_{23}}{|d_{23}|^3} \right). \end{aligned}$$

They preserve $2|z|^2 - (m+n) - e|w|^2 = 0$ and are analytic at $w = 0$ when the other two separations are positive. At $v = 1$ the launch is $(w, z, e, G, P) = (1, 0, -\sqrt{2}, i/2, 0)$; its only nonzero parameter derivatives are $m_v = -1/2$, $n_v = 1/2$, and $G_{x,v} = 1/(2\sqrt{2})$.

6.1 The initial arc and collision unfolding

The first syzygy, or collinear configuration, is reached without a brake for all tied parameters sufficiently near $v = 1$. The relevant variational argument also establishes the torque inequality used in Section 9. Write $F_0(v, t) = \eta(v, t) - h(v, t)$ for the history-threshold difference defined there, and let $\tau(v)$ be the first syzygy time. On the symmetric orbit, $F_0(1, t) = 0$. Differentiation of the Newtonian equations and the exact tied launch gives an analytic function $g(t) = \partial_v F_0(1, t)$ with

$$g(0) = \frac{32 + 8\sqrt{2}}{7} > 0.$$

A validated orbit-and-tangent cover proves $g > 6$ up to $t = 10^{-3}$, $g > 4$ on $[10^{-3}, 0.43]$, and $g_t < -6$ from 0.43 through the unique transverse syzygy. The collinear identity for the torque threshold gives $F_0(v, \tau(v)) = 0$, hence $g(\tau(1)) = 0$. Analytic division yields

$$F_0(v, t) = (v - 1)(t - \tau(v)) F_2(v, t), \qquad F_2(1, \tau(1)) = g_t(\tau(1)) < 0.$$

This factorization controls the terminal part of the arc, while compact continuity controls its interior. Thus $F_0 < 0$ for $v < 1$ sufficiently close to one and $0 < t < \tau(v)$. The same tangent cover proves $r_{12} > r_{23} > r_{31}$ there. A pair angular momentum then remains nonzero through syzygy, excluding a brake. The launch division and orbit-and-tangent equations are checked by exact symbolic regression; the full interval inputs are indexed in Appendix A.

At the first endpoint collision, an oriented C^1 Poincaré calculation gives $z_r < -4/5$ and $\partial_v w_i < -30$. Thus the map $(\sigma, v) \mapsto (w_r, w_i)$ has nonsingular derivative there. The nearby members miss that collision, with

$$\min r_{12} = \chi^2(1-v)^2 + o((1-v)^2), \qquad \chi^2 > 900.$$

At the section $w_r = 0$, $\ell_{12} = -2w_i z_r > 0$ for $v < 1$ sufficiently close to one, whereas its launch sign is negative. Thus the launch sign has already reversed by this section; it cannot be assumed to persist through later encounters.

6.2 Continuation to a terminal escape section

Four comparison collisions must be controlled before the escape criterion applies. Their regularized transversality and the nonzero brake residual give a single punctured neighborhood on which the entire argument holds.

Theorem 18 (Punctured near-isosceles nonperiodicity)**.** *There exists $v_* < 1$ such that every real tied member with $v_* < v < 1$ is nonperiodic. It is collision-free and has no labelled brake through a fixed LC section; thereafter it either ends in a classical binary collision or has body 3 escape. Consequently the same conclusion holds for infinitely many primitive Pythagorean triples.*

Proof. The exact LC energy constraint is

$$2|z|^2 - (m+n) - e|w|^2 = 0.$$

On the symmetric endpoint orbit it gives $2|z|^2 = \sqrt{2}$ whenever $w = 0$, so every selected collision is transverse. Pinned oriented C^1 Poincare maps enclose four such zeros before $\sigma = 7$ and put the fifth strictly after 7. At each of the first four zeros, the tied normal derivative $(w_i)_v$ is bounded away from zero. Thus every Jacobian

$$\frac{\partial(w_r, w_i)}{\partial(\sigma, v)}$$

is nonsingular. The inverse-function theorem isolates all four zeros, while a 7000-step C^0 cover proves

$$r_{31}^2, r_{23}^2 > \frac{1}{500} \qquad (0 \le \sigma \le 7).$$

Compactness outside the four collision boxes now gives a punctured neighborhood in which every nearby tied orbit is collision-free through $\sigma = 7$.

At an ordinary point of this LC chart,

$$\dot{g} = \frac{2z}{\bar{w}}, \qquad \dot{G} = P.$$

Hence, after center-of-mass reduction, a labelled brake is equivalent to $z = P = 0$. The polynomial residual $\mathcal{R}_{\mathrm{LC}} = |z|^2 + |P|^2$ remains regular at comparison collisions. At the $\sigma = 1/2$ interface the cover gives $G_y > 1/10$, so the unique first syzygy lies later. Every complete interval tube from that overlapping pre-syzygy interface to $\sigma = 7$ satisfies $\mathcal{R}_{\mathrm{LC}} > 2$. Compact continuity transfers this inequality to the nearby classical orbits; the initial-arc result described above covers the initial segment.

At $\sigma = 7$, take $\eta = 4$ in Lemma 6. The same interval cover proves, with $M = m + n = \sqrt{2}$, $R = M/4$ and $s = |G| - R$,

$$s > 1, \qquad \dot\rho > 2, \qquad \delta = \frac{\dot\rho^2}{2} - \frac{M+1}{s} > \frac{1}{20},$$

and

$$-4 - e - \frac{\sqrt{2MR}}{\sqrt{2\delta}\, s^2} > 2.$$

All inequalities are strict and therefore persist in a tied neighborhood. The lemma says that each future trajectory either suffers an inner classical collision or escapes with uniformly positive outward radial speed. Neither possibility permits a later brake. Rational Euclid parameters are dense in the resulting open u interval, giving infinitely many primitive triples. □

The cutoff v_* is existential. No numerical width is claimed for this endpoint neighborhood. An effective bound would identify a corresponding explicit range of integer triangles.

7 Thin triangles and escape windows

At the thin end of the family, the small mass $B = \epsilon$ is far from a tight pair of masses $A = \sqrt{1-\epsilon^2}$ and 1, initially separated by ϵ. Two clocks then coexist: the inner pair moves on a time scale $\epsilon^{3/2}$, while the outer body returns on a time scale of order one. A single outer excursion contains an unbounded number of binary cycles as $\epsilon \to 0$. Exclusion of one early encounter therefore leaves an increasing number of later passages to control.

7.1 The first close encounter

For $X = q_3 - q_1$, put $X = \epsilon x$ and $t = \epsilon^{3/2}\tau/\sqrt{A+1}$, and rotate so that $x(0) = 1$. The leading motion is radial Kepler fall. In Levi–Civita coordinates $x = z^2$, $d\tau = |z|^2 ds$, its solution is $z_0(s) = \cos(s/\sqrt{2})$, with collision at $s_c = \pi/\sqrt{2}$. The true parameter-dependent equations are analytic at this comparison collision. Their first transverse variation appears only at order five:

$$v_{ss} + \tfrac{1}{2}v = -\tfrac{3}{8}\cos^5(s/\sqrt{2}), \qquad v(0) = v_s(0) = 0, \qquad v(s_c) = -\frac{15\pi}{128}.$$

Consequently the physical first encounter misses collision by

$$r_{13,\min} = \frac{225\pi^2}{16384}\epsilon^{11}(1 + O(\epsilon)). \tag{5}$$

The proof uses the simple zero of $\operatorname{Re} z_0$ and the nonzero fifth-order imaginary displacement; away from that zero compact continuity keeps all pairs separated. Squaring the regularized displacement and restoring the factor ϵ gives the eleventh power. The symbolic coefficient is also obtained directly from the oscillator Green function: with $a = s/\sqrt{2}$,

$$v(s_c) = -\frac{3}{4}\int_0^{\pi/2} \cos^6 a\, da = -\frac{15\pi}{128}.$$

The same calculation gives the specific pair angular momentum $\ell_{13} = -(15\pi/64)\epsilon^{11/2} + O(\epsilon^{13/2})$.

The first passage is not an escape certificate. Its outer radial escape margin is $-2 + O(\epsilon)$, while the pair remains tightly bound. The subsequent return must be analyzed.

7.2 The restricted problem at the outer plunge

Let $X = q_3 - q_1$ and $Y = q_2 - (Aq_1 + q_3)/(A+1)$. Near the outer plunge, set $X = \epsilon R$, $Y = \epsilon Z$, and $t = t_* + \epsilon^{3/2}\theta$. With $\Phi(v) = v/|v|^3$, the limiting equations are

$$R_{\theta\theta} = -2\Phi(R), \qquad Z_{\theta\theta} = -\Phi(Z + R/2) - \Phi(Z - R/2). \tag{6}$$

The two heavy masses have become equal, and the light particle no longer affects their motion. On the invariant symmetric subspace, $R = (r, 0)$, $Z = (0, z)$, and

$$r = \cos^2\psi, \qquad d\theta = r\,d\psi, \qquad z_{\theta\theta} = -\frac{2z}{(z^2 + r^2/4)^{3/2}}.$$

Here the binary has apocenter one and is continued through its collisions in Levi–Civita coordinates. All limiting arguments through these collisions require the light particle to remain separated.

Release the light particle from $z = 0$ at binary apocenter with speed v. Small positive speeds turn, whereas sufficiently large speeds escape. The boundary of the turning set contains a parabolic orbit, tending to infinity with zero speed; reflection and time reversal extend it to a parabolic-to-parabolic orbit. For $z > 0$, $z_{\theta\theta} < 0$ prevents a bounded nonturning boundary orbit. Openness of finite turning and positive-speed escape excludes both alternatives at the boundary. The speed v_* satisfies $\sqrt{8} < v_* < 4$; interval integration with the analytic tail bound

$$0 \le H_K(\infty) - H_K(z) \le \frac{1}{4z^3}, \qquad H_K = \frac{z_\theta^2}{2} - \frac{2}{z},$$

refines this to $2.9051113 < v_* < 2.9051116$.

7.3 Matching the binary clock and the full incoming state

Write $M = A + 1$, $N = M + \epsilon$, and $\rho_0^2 = 2A^2/M$. Replacing the inner pair by its combined point mass provides an explicit reference phase:

$$\Phi_{\mathrm{ref}}(\epsilon) = \frac{\pi}{\epsilon^{3/2}}\sqrt{\frac{M}{N}}\,\rho_0^{3/2} = \pi\epsilon^{-3/2} - \frac{\pi}{4}\epsilon^{-1/2} - \frac{15\pi}{32}\epsilon^{1/2} + O(\epsilon^{3/2}). \tag{7}$$

The reference clock approximates the asymptotic phase intercept of the interacting system. To transfer an escape window, one needs convergence of the full incoming state as well as its phase.

Theorem 19 (Incoming-state matching)**.** *On a fixed sufficiently large scaled incoming section Σ_K^-, let $\Gamma_K^-(\chi)$ denote the incoming parabolic state of the equal-heavy-mass restricted problem, parameterized by its asymptotic mean-anomaly intercept χ. As $\epsilon \to 0$, either a classical collision ends the exact tied orbit before it reaches the section, or its full regularized section state $S_\epsilon(K)$ satisfies*

$$\mathrm{dist}\big(S_\epsilon(K), \Gamma_K^-(\Phi_{\mathrm{ref}}(\epsilon))\big) = o(1).$$

Proof. Stop the outer infall at physical radius $\rho = \epsilon^\alpha$ with $0 < \alpha < 1/6$. Cancellation of the outer dipole gives a monopole-force remainder $O(\epsilon^2\rho^{-4})$. Inner-energy control, variation of the regularized oscillator, and comparison of its physical clock then give an incoming intercept error $O(\epsilon^{1/2-3\alpha}) = o(1)$. On the remaining tail, write $y = |Z|$, $y_e = \epsilon^{\alpha-1}$, $\mathcal{H}_N = |Z_\theta|^2/2 - N/y$, and $\mathcal{L} = Z \times Z_\theta$. The dipole cancellation and a simultaneous energy and angular-momentum bootstrap give, uniformly for $K \le y \le y_e$,

$$|\mathcal{H}_N - E_{\mathrm{ref},\epsilon}| \le Cy^{-3}, \qquad |\mathcal{L}| \le Cy^{-3/2}, \qquad -y_\theta \asymp y^{-1/2},$$

where $E_{\mathrm{ref},\epsilon} = -N\epsilon/\rho_0$ is the scaled monopole energy. Binary control through every close passage uses the oscillator invariants

$$\mathcal{J} = (E+1, L_b, K_b), \quad L_b = z_x p_y - z_y p_x, \quad K_b = p_x p_y - (E/2) z_x z_y,$$

with $p = z_s$. On the radial Kepler circle the normal Jacobian $\det \partial(L_b, K_b)/\partial(z_y, p_y) = -1/2$ stays nonzero, including at binary collision. The forced equations give $|\mathcal{J}_s| \le C\epsilon y^{-3}$; summing complete LC blocks and bounding the two incomplete end blocks gives a clock error $O(\epsilon^{1-3\alpha/2}) + O(\epsilon|\log \epsilon|) = o(1)$.

Retain the finite-ϵ monopole flight-time subtraction until y is fixed. Its phase error is at most $Cy^{-1/2} + o(1)$, which includes the finite quadrupole correction from infinity. Compactness in LC variables gives a limiting restricted tail. Its transverse displacement has the inherited bound $O(y^{-1})$, whereas the homogeneous transverse modes grow as y and $y^{1/2}$. Thus the limit is rectilinear. Taking $\epsilon \to 0$ at fixed y and then $y \to \infty$ identifies the intercept as $\Phi_{\mathrm{ref}}(\epsilon)$ in the limiting sense. Uniqueness of the incoming parabolic graph gives the stated full-state convergence. The uniform tail estimates and their LC block bounds are recorded in `docs/INCOMING_TAIL.md`. □

7.4 Transversality and the escape arcs

At $z = 0$, let $V_u(\phi)$ and $V_s(\phi)$ be the positive crossing speeds on the incoming and outgoing parabolic curves, with $\phi = 0$ at binary apocenter. Reversibility gives $V_s(\phi) = V_u(-\phi)$. The transversality condition is $V_u'(0) \ne 0$. Fixing the phase is essential: the time-translation Jacobi field decays at parabolic infinity and is not removed merely by prescribing zero limiting data.

For the longitudinal Jacobi field h, put $q = h/v$, $w = z_\theta$, $p = q_\theta$, and $d^2 = z^2 + r^2/4$. The finite regular system is

$$z_\psi = rw, \quad w_\psi = -\frac{2rz}{d^3}, \quad q_\psi = rp, \quad p_\psi = r\frac{4z^2 - r^2/2}{d^5} q, \qquad (z, w, q, p)(0) = (0, v, 1, 0).$$

An order-20 interval Taylor calculation on 256 overlapping speed slabs proves

$$p(\pi/2) > 1/125 \qquad \text{uniformly for } 14/5 \le v \le 4.$$

Here v denotes launch speed, not the near-isosceles parameter. Force comparison gives $h > 0$ through this first contraction and a positive Jacobi coefficient thereafter. Thus the certified derivative forces $h(\theta)$ to grow at least linearly.

To relate this growth to transversality, compactify infinity by $z = 2/x^2$. The exact field is analytic at $x = 0$, and its time-π map in binary eccentric anomaly, after $X = x$, $Y = w - \sqrt{2}x$, is

$$X_1 = X - \frac{\pi}{8} X^3 (Y + \sqrt{2} X) + O_7, \qquad Y_1 = Y + \frac{\pi\sqrt{2}}{8} X^3 Y + O_7.$$

The degree-four terms contract the positive X axis and expand its normal direction; O_7 denotes analytic terms of total degree at least seven. McGehee's degenerate stable-manifold theorem gives a parabolic graph $Y = \varphi(X)$ with $\varphi'(X) \to 0$ [21]. Along it, $X_n^{-3} \asymp n$ and every tangent tends to zero. If the parabolic curves were tangent at $\phi = 0$, reversibility would make both slopes vanish. The difference of the time-shift tangent and the parabolic tangent with fixed phase would then be the field h above. But $\delta X_n = -X_n^3 h(\theta_n)/4$ would stay bounded away from zero, contradicting tangent contraction. This proves transversality.

Consequently $V_u - V_s$ changes sign linearly at zero. The parabolic graph separates an open hyperbolic-escape side from an open finite-turn side. This is the source of the escape-intercept arcs used below.

7.5 Transfer to integer triangles

Fix a compact subarc J of positive length strictly inside a restricted escape window. Incoming-state matching and continuous dependence carry sufficiently small exact tied members with $\Phi_{\mathrm{ref}}(\epsilon) \bmod 2\pi \in J$ to a finite outgoing section. There the binary specific energy and outer speed have the forms

$$e = -C/\epsilon + o(\epsilon^{-1}), \qquad \dot{\rho} = s\epsilon^{-1/2} + o(\epsilon^{-1/2}), \qquad C, s > 0,$$

with strict margins uniform on the compact arc. At a sufficiently distant scaled section, Lemma 6 applies. An earlier classical collision already excludes periodicity; every other member escapes or subsequently collides. This proves all-time exclusion on these windows, not just on the first excursion.

Theorem 20 (Infinitely many thin-triangle intervals)**.** *There are infinitely many open intervals of real u accumulating at zero on which the exact tied solution is nonperiodic. Each contains rational parameters and therefore primitive integer Pythagorean triples.*

Proof. Equation (7) gives $\Phi'_{\mathrm{ref}}(\epsilon) \sim -(3\pi/2)\epsilon^{-5/2}$. The preimages of the interior of J therefore include infinitely many disjoint intervals accumulating at zero. The uniform transfer above applies to every sufficiently small one. Finally $B(u)$ is a homeomorphism near zero and rational u are dense. Different parameters in the fundamental interval give different primitive triples after scaling. $\square$

Theorem 21 (A positive-density explicit family)**.** *For a set of positive lower natural density of integers $n \geq 1$, the primitive triple*

$$(a_n, b_n, c_n) = (4n^2 - 1, 4n, 4n^2 + 1)$$

has a nonperiodic Pythagorean–Burrau solution.

Proof. Here $u_n = 1/(2n)$ and $B_n = 4n/(4n^2+1)$. The exact reference clock gives

$$F(x) := \frac{\Phi_{\mathrm{ref}}(B_x)}{2\pi} = \frac{(2x-1)^{3/2}(2x+1)(4x^2+1)^{3/4}}{32x^{5/2}},$$

with

$$F''(x) = \frac{3}{8}x^{-1/2} + \frac{1}{32}x^{-3/2} + O(x^{-5/2}).$$

The differentiated remainder follows from analyticity in $1/x$ after factoring out $x^{3/2}$. For each nonzero integer h, van der Corput's second-derivative estimate bounds the Weyl sum on $N < n \leq 2N$ by $O_h(N^{3/4})$. Dyadic summation and Weyl's criterion give equidistribution of $F(n)$ modulo one. Thus the compact arc J selects indices of density $|J|/(2\pi) > 0$, and the escape transfer proves nonperiodicity for every sufficiently large selected index. Since $\gcd(4n, 4n^2 - 1) = 1$, the triples are primitive. This density concerns the displayed family of indices, not the set of all primitive Pythagorean triples ordered by hypotenuse. $\square$

The returning side requires additional control through successive central encounters. The next section gives the available first-turn exclusion, the near-triple-collision reduction, and the local collision classifications.

8 Returning trajectories and collision boundaries

The escape windows leave an infinite family of returning trajectories. Two further results restrict their possible brakes: a uniform angular-momentum estimate at the first outer turn, and a local analysis of passages near triple collision. Throughout this section, a collision of a positive-mass trajectory ends its classical solution. Continuation through collision is used only in the auxiliary regularized equations.

8.1 The first outer turn

On the returning side of the restricted separatrix, the first-turn time $T(\phi)$ tends to infinity as $\phi \to 0$. Since the binary mean anomaly is $\phi+4\theta$, continuity supplies phases $\phi_n \to 0$ with $\phi_n+4T(\phi_n) = 2\pi n$. At these phases the outer turn coincides with a binary apocenter. Radial Kepler comparison gives

$$T_n = \frac{\pi n}{2} + o(1), \qquad Z_{\mathrm{t},n} = (2n)^{2/3}(1 + o(1)).$$

These are brakes of the regularized massless problem, so zeroth-order convergence alone cannot exclude brakes of the full system.

The first transverse variation supplies the obstruction. On the centered parabolic orbit of Section 7.2, let p_- solve

$$p_-'' = \frac{r^2 - 2z^2}{d^5} p_-, \qquad p_- - z = O(|z|^{-1}) \quad (\theta \to -\infty), \qquad d^2 = z^2 + r^2/4,$$

where primes denote θ derivatives. Its Wronskian with z satisfies

$$(zp_-' - z'p_-)' = \frac{3r^2 z p_-}{2d^5}, \qquad \mathcal{W}_\infty = \lim_{\theta\to\infty} (zp_-' - z'p_-).$$

An outgoing solution k_+, normalized by $k_+/z \to 1$ and $zk_+' - z'k_+ \to 0$, gives by time reversal $\mathcal{W}_\infty = 2k_+(0)k_+'(0)$. Analytic Volterra bounds attach this solution at finite radius: for $z \geq K \geq 1$,

$$1 \leq \frac{k_+}{z} \leq Q_K = \frac{1}{1 - 1/(7K^2)}, \qquad -\frac{Q_K}{\sqrt{7/2}\, K^{3/2}} \leq zk_+' - z'k_+ \leq 0.$$

Interval propagation of these bounds gives

$$k_+(0) > 7/20, \qquad k_+'(0) > 3/5, \qquad 11/25 < \mathcal{W}_\infty < 9/20.$$

Theorem 22 (Uniform first-turn exclusion)**.** *For exact tied trajectories with $B \to 0$ whose restricted intercepts approach the centered parabolic phase from the returning side, either a prior classical collision occurs or, at the first outer turn,*

$$\frac{Y \times \dot{Y}}{B^{3/2}} \longrightarrow \frac{\mathcal{W}_\infty}{2} > \frac{21}{100}.$$

The convergence is uniform in turn height. Hence no second brake occurs at these first turns. On return to a large fixed scaled radius K, the quotient differs from $\mathcal{W}_\infty/2$ by $O(K^{-3/2}) + o(1)$ as $B \to 0$.

Proof. Set $\Lambda_B = (Z \times Z')/B$, $M = A + 1$, and $N = M + B$. The exact torque factorization is

$$\Lambda'_B = -\frac{NA}{BM^2}(Z \times R)\big(|Z + R/M|^{-3} - |Z - AR/M|^{-3}\big).$$

The squared denominators differ by $2Z \cdot R + B^2|R|^2/M^2$. First-order incoming matching and the divided LC normal equations retain $|Z \cdot R| \leq CB|Z|$ on the outgoing tail. A simultaneous binary-energy bootstrap keeps the LC oscillator bounded. Writing $y = |Z|$, radial comparison gives $-C/y^2 \leq y'' \leq -c/y^2$ and $\int y^{-3}\, d\theta \leq CK^{-3/2}$ from $y = K$ to any first turn. This estimate includes the neighborhood where $y' = 0$ and is uniform in turn height. It bounds the change in Λ_B by $CK^{-3/2}$. At fixed K, the divided transverse matching gives $\Lambda_B \to (zp'_- - z'p_-)/2$. Letting $K \to \infty$ proves the claim. The same radial and torque estimates hold on the returning leg. □

For each fixed late restricted resonance, the transverse velocity is also nonzero at small positive mass, but its coefficient tends to zero as the resonance index increases. The angular-momentum estimate above provides the uniform exclusion. It leaves the next central encounter uncontrolled. At its centered parabolic endpoint, the asymptotic coefficient $\gamma = \lim p_-/z$ satisfies $-1/100 < \gamma < -1/250$, and the outgoing Wronskian after the second encounter is $-2\gamma\mathcal{W}_\infty > 21/6250$; the interior returning phases still require classification.

8.2 The triple-collision endpoint and its joint scaling

Another boundary of the returning region approaches equilateral triple collision. The longitudinal and transverse unstable exponents in logarithmic size are

$$\mu = \frac{1+\sqrt{19}}{4}, \qquad \tau = \frac{1+\sqrt{7}}{4}, \qquad 1 < p := \mu/\tau < 2.$$

Let H_B, T_B be the corresponding normalized unstable amplitudes after subtracting the local collision-stable graph. The odd transverse coefficient satisfies $T_B/B \to \Theta_* \neq 0$. Define

$$\rho_T = |T_B|^{1/\tau}, \qquad \kappa_B = \frac{H_B}{|T_B|^p}.$$

Scaling size by ρ_T and time by $\rho_T^{3/2}$ produces a planar massless limit indexed by $\kappa = \lim \kappa_B$. Its vector equations are (6), now with a zero-energy radial binary. The longitudinal scaling gives the rectilinear exits when $\kappa_B \to \pm\infty$.

The nonzero coefficient Θ_* is fixed by a certified transverse intersection of the incoming parabolic curve with the collision-stable curve. The latter has a unique connection in its normalized amplitude bracket $[-0.24696, -0.24694]$. Along the terminal binary infall, the canonical transverse collision modes have powers $r^{\beta_\pm}$, $\beta_\pm = (3 \pm \sqrt{7})/4$, and Wronskian $-\sqrt{7}$. An interval bound $W(J, P_+) < -1/2$ for the returned field J therefore proves its slow coefficient $A_- = W(J, P_+)/(-\sqrt{7}) > 0$. A second finite certificate, attached to analytic incoming bounds, proves that the universal slow mode has positive first-turn transfer $K_- > 0$. These signs exclude a brake near the second turn on the selected local restricted branch.

For a finite light mass this conclusion needs a scale condition. If $\varepsilon = (\pi - \phi)/4$ is the restricted center-to-collision time gap and $\omega_0(B) \to 0$ is the fixed-section state-matching error, the transfer holds when

$$\omega_0(B) + B = o(\varepsilon^{(1+\sqrt{19})/6}).$$

The transverse displacement relative to the inner size is then $B\varepsilon^{-(1+\sqrt{7})/6} = o(1)$. The complementary layer is described by the planar κ family. It cannot contain a brake *during* a sufficiently small near-triple passage: energy at a brake requires the exact inequality

$$|R| > \frac{A}{BU_0} = \frac{1-B^2}{1+B^2-B^4} = 1 - 2B^2 + O(B^4).$$

Possible brakes after re-expansion remain the issue.

8.3 Planar collision boundaries and escape neighborhoods

On a radial half of the limiting binary, set $R = 9^{1/3}t^{2/3}e_x$, $Z = 9^{1/3}t^{2/3}w$, and $\zeta = \log t$. Then

$$w_{\zeta\zeta} + \tfrac{1}{3}w_\zeta = \nabla W(w), \qquad W(w) = \tfrac{1}{9}\big(|w|^2 + |w+e_x/2|^{-1} + |w-e_x/2|^{-1}\big).$$

The shape energy obeys $\mathcal{E}_\zeta = -|w_\zeta|^2/3$, where $\mathcal{E} = |w_\zeta|^2/2 - W$. For $d_\pm = |w \pm e_x/2|$,

$$9W = \sum_\pm (d_\pm^2/2 + d_\pm^{-1}) - 1/4 \geq 11/4,$$

with equality only at the equilateral shapes. A nonconstant orbit tending to an equilateral shape has $\mathcal{E} > -11/36$ at finite times and therefore cannot have zero shape velocity there.

Normalize the lower-equilateral stable family by $w = (0, -\sqrt{3}/2) + P(a,b)$, where $P(a,b) = (a,b) + O_2$, $a = -e^{-2\tau\zeta/3}$, and $b = \kappa e^{-2\mu\zeta/3}$. These choices fix the scale and the meaning of κ in the numerical enclosures. Substitution in the shape equation determines its quartic jet P_4 recursively. An exact homological inverse bound 30/17 and a contraction factor below 1/10 give $\|P - P_4\|_{1/200,1/2500} < 1/125000000$ in the weighted coefficient ℓ^1 norm. Restriction to smaller polydiscs bounds both the state and its κ derivative. Higher jets improve the later collision enclosures.

In a selected light–primary chart $q = u^2$, $dt = |u|^2 d\sigma$, the equation $q_{tt} = -\Phi(q) + F(q,t)$ becomes

$$u_\sigma = v, \qquad v_\sigma = (h/2)u + (|u|^2\bar{u}/2)F, \qquad h_\sigma = 2\,\mathrm{Re}(uv\bar{F}),$$

with $2|v|^2 - 1 - h|u|^2 = 0$. The other primary must remain separated. The square collision map $(\kappa,\sigma) \mapsto (\Re u, \Im u)$ is tested by interval Newton. Four local roots have been enclosed:

Encounter chart	Primary	Enclosure for κ
1	positive	$[1.2679351752, 1.2679351759]$
2	negative	$[1.2640119191, 1.2640119311]$
3	positive	$[1.2640090993171269, 1.2640090999643190]$
4	negative	$[1.26400909893331527, 1.26400909893331813]$

Uniqueness is local to the parameter-time boxes in the certificates. Earlier collisions are excluded for each listed root. At the first root, $D_c = \det(\partial_\kappa u, v) \in (-3.52212, -3.38401)$; projection onto $v^\perp$ gives

$$r_{\mathrm{min,local}} = 2D_c^2(\kappa - \kappa_c)^2 + O(|\kappa - \kappa_c|^3).$$

For positive mass, a colliding-pair center coordinate cancels the singular force from the complementary equation. This gives a common analytic LC chart, with $|v|^2 = (1+B)/2$ at the first selected collision. Differentiated inclination through the logarithmic saddle dwell then supplies a local collision graph $\kappa_c(B,T,S,\Xi)$, where S is the stable entrance coordinate and Ξ collects the remaining energy and clock deviations; $\widehat{\Xi}$ denotes their Newtonian-scaled values. The graph converges to the limiting root when

$$B|\log|T|| + B + |T|^{2-p} + e^{-cL_T} + \|\widehat{\Xi}\| \to 0, \qquad L_T = -\tau^{-1}\log|T|.$$

The differentiated contraction uses a fixed weight $\mu/2 < \omega < \tau$; thus the quadratic Green denominator $2\omega - \mu$ stays positive. This is the step that justifies derivative convergence through the increasingly long saddle passage.

Finite regularized chart chains classify neighborhoods of the first, second, and fourth roots as collision or escape. For example, ten adjacent interval propagations cover $[1.2679251755, 1.2679451755]$ around the first root. They reach an outgoing section with the strict margin

$$\dot{\rho}_0 - \frac{2}{(3/2)(\rho_0 - r_0)} - \sqrt{4/r_0 + 2/100} - \frac{3}{2} > 0.65.$$

This is Lemma 7 with $\varepsilon = 1/100$, $c = 3/2$, and limiting inner energy zero. The scaled tidal allowance is $O(B)$, so the strict margin transfers to sufficiently small compatible positive-mass states. The second-root limiting box $[1.2640099161, 1.2640126461]$ has the same local finite-mass conclusion. The fourth-root chain requires simultaneous two-centre regularization to resolve subsequent close passages; its finite-mass-form escape margin exceeds 10.39. A separate 131-tile limiting cover proves collision or escape on $[1.264009099014, 1.264009099457]$; that explicit interval is a result for the massless family. None of these local results classifies every component of the κ family.

8.4 Exact collisions on the tied curve

Theorem 23 (Real collision parameters)**.** *Infinitely many distinct real tied parameters $B_j \downarrow 0$ have a finite-time classical collision.*

Proof. Each sufficiently late winding of the reference phase gives a disjoint parameter bracket crossing the two sides of the certified triple endpoint. An earlier collision anywhere in a bracket already supplies the assertion. Otherwise, incoming matching transfers opposite signs of H_B to its endpoints. Within that bracket choose a segment from $H_B = 0$ to $H_B = B^{p/2}$ on which $0 \le H_B \le B^{p/2}$. Local uniqueness of the restricted intersection localizes this shrinking segment at the endpoint, uniformly giving $T_B/B \to \Theta_* \neq 0$ and the collision-graph hypotheses. The continuous function

$$G(B) = H_B - |T_B|^p \kappa_c(B, T_B, S_B, \Xi_B)$$

is negative at the first endpoint and positive at the second, since $|T_B|^p = O(B^p) = o(B^{p/2})$. A zero is an exact light–heavy collision. The disjoint brackets accumulate at zero. $\square$

The roots are not known to be rational. On a bracket without an earlier collision, the same normalized coordinate $H_B/|T_B|^p$ traverses the first collision-or-escape interval. Its interior preimage supplies an additional nonempty open tied window of nonperiodicity. Classification of all remaining returning components, including their later turns, is still open.

9 Torque history and the remaining analytic obstruction

An analytic restriction on trajectories released from the tied brake curve could extend the interval results. Integrating the pair torques gives necessary cancellation conditions for a second brake and a scalar equation for the accumulated torque ratio. The proposed global sign consequence remains unproved.

9.1 Integral constraints

Let $d_{ij} = q_j - q_i$, $r_{ij} = |d_{ij}|$, and $\Delta_2 = (q_2 - q_1) \times (q_3 - q_1)$ be twice the signed area. Define the specific pair angular momenta $\ell_{ij} = d_{ij} \times \dot{d}_{ij}$, using the cyclic pairs $12, 23, 31$. Direct differentiation gives

$$\begin{aligned} \dot{\ell}_{12} &= m_3 \Delta_2 (r_{23}^{-3} - r_{31}^{-3}), \\ \dot{\ell}_{23} &= m_1 \Delta_2 (r_{31}^{-3} - r_{12}^{-3}), \\ \dot{\ell}_{31} &= m_2 \Delta_2 (r_{12}^{-3} - r_{23}^{-3}). \end{aligned} \tag{8}$$

Their derivatives have signs $(-, +, -)$ at launch in the strict fundamental interval. Since every ℓ_{ij} vanishes at a brake, each right-hand side must integrate to zero before a second brake. Its initial strict sign must therefore be reversed somewhere. Before the first syzygy, this requires reversals of all three initial side comparisons, at possibly different times. It does not prove that those reversals cannot occur.

Integrating Lagrange–Jacobi between two brakes also gives

$$\int_0^\tau U \, dt = 2U_0 \tau, \qquad \int_0^\tau K \, dt = U_0 \tau.$$

Integrating the labelled accelerations against the fixed launch positions also gives

$$\int_0^\tau \sum_{i<j} m_i m_j \frac{d_{ij}(0) \cdot d_{ij}(t)}{r_{ij}(t)^3} \, dt = 0.$$

The integrand is initially positive; its later sign is not controlled.

9.2 The torque ratio and its threshold

On the arc from launch to the first loss of positive area or strict ordering $r_{12} > r_{23} > r_{31}$, put $R = r_{12}$, $x = r_{23}/R$, $y = r_{31}/R$, and define

$$F = \Delta_2 (r_{31}^{-3} - R^{-3}) > 0, \qquad G = \Delta_2 (r_{23}^{-3} - R^{-3}) > 0, \qquad k = \frac{G}{F} = \frac{y^3(1 - x^3)}{x^3(1 - y^3)}.$$

Integrating the torques from release gives

$$\eta = \frac{\int_0^t G \, d\tau}{\int_0^t F \, d\tau} = \frac{m_1}{m_2} \frac{-\ell_{31}}{\ell_{23}}, \qquad \dot{\eta} = \frac{F}{\int_0^t F \, d\tau} (k - \eta). \tag{9}$$

Thus η is a weighted average of the preceding values of k. Strict decrease of k would imply $k(t) < \eta(t) < k(0)$; establishing that decrease along the tied trajectories is the remaining problem.

Use shape time $ds = dt/R^{3/2}$ and write $W = \ell_{23}/\sqrt{R}$, $\delta = \Delta_2/R^2$. Eliminating the two shape velocities gives the rational identity

$$(\log k)_s = \frac{W}{\delta} \mathcal{A}(u, x, y)(\eta - h(u, x, y)), \qquad \mathcal{A} > 0, \quad h > k. \tag{10}$$

For an explicit definition, put $m = m_1$, $n = m_2$, and

$$\begin{aligned} \mathsf{B} &= n(-x^5 - x^3 y^2 + x^3 + x^2 + 2y^5 - y^2 - 1) \\ &\quad + x^3 y^2 (x^2 - y^2 + 1) + y^5 (-x^2 + y^2 + 1) - 2y^2, \\ \mathsf{N} &= m(2x^5 - x^2 y^3 - x^2 - y^5 + y^3 + y^2 - 1) \\ &\quad + x^5 (x^2 - y^2 + 1) + x^2 y^3 (-x^2 + y^2 + 1) - 2x^2, \\ h &= \frac{y^2 \mathsf{N}}{x^2 \mathsf{B}}, \qquad \mathcal{A} = \frac{-3\mathsf{B}}{2my^2(x-1)(y-1)(x^2+x+1)(y^2+y+1)}. \end{aligned}$$

Here h denotes the torque threshold. The ordered-triangle substitution $x = 1-ab/2$, $y = 1-a+ab/2$, $u = (\sqrt{2}-1)v$, with $0 < a, b, v < 1$, maps the domain to a cube. After clearing denominators with known sign, exact tensor-Bernstein coefficients prove $\mathcal{A} > 0$ and $h - k > 0$. Thus (10) reduces the desired monotonicity to $\eta < h$. Section 6 proves this inequality on the first arc near the equal-leg endpoint.

9.3 A corrected conditional sign-change argument

Define the gap, lag, and threshold distance

$$d = h - k > 0, \qquad e = \eta - k, \qquad w = h - \eta = d - e,$$

and the positive coefficients

$$\lambda = \frac{m_1 \delta (y^{-3} - 1)}{W}, \qquad c = \frac{kW\mathcal{A}}{\delta}.$$

Equations (9)–(10) give

$$e_s + (\lambda + c)e = cd, \qquad w_s + (\lambda + c)w = f, \qquad f = d_s + \lambda d. \tag{11}$$

For s strictly before syzygy, the launch condition has the improper integral representation

$$e(s) = \lim_{\varepsilon \downarrow 0} \int_\varepsilon^s c(\tau)d(\tau) \exp\left[-\int_\tau^s (\lambda + c)\, d\xi\right] d\tau.$$

At launch W has a simple zero, $\lambda \sim 1/s$, and $e(0) = 0$ after analytic extension, so the homogeneous launch contribution tends to zero. For a fixed interior reference point $s_0 > 0$, define the integrating factor

$$M(s) = \exp \int_{s_0}^s (\lambda + c)\, d\xi, \qquad (Mw)_s = Mf.$$

At the first transverse ordered syzygy $s = S$, collinear momentum constraints force $w(S) = 0$, while $\delta \to 0$ and $c = kW\mathcal{A}/\delta$ can diverge. The endpoint equality therefore does not imply $Mw \to 0$, as assumed in an earlier draft.

Lemma 24 (Sufficient sign-change criterion with terminal control)**.** *Assume $w > 0$ near launch, f is positive up to one interior sign change and negative thereafter, and*

$$\liminf_{s \uparrow S} M(s)w(s) \geq 0. \tag{12}$$

Then $w > 0$ throughout $0 < s < S$.

Proof. Start at any sufficiently small positive time where $w > 0$. Since $M > 0$, the product Mw increases until the switch and then strictly decreases. If it reached zero before S, its terminal lower limit would be negative, contrary to (12). □

The missing hypothesis cannot be omitted from this scalar argument. On $0 < s < 1$ take

$$a = \frac{1}{s} + \frac{1}{1-s}, \qquad w = (1-s)(1+s-3s^2), \qquad f = (1-s)\left(\frac{1}{s} + 2 - 9s\right).$$

Then $w_s + aw = f$, $w(0) = 1$, and $w(1) = 0$. The forcing changes from positive to negative exactly once, at $(1+\sqrt{10})/9$, yet w becomes negative after $(1+\sqrt{13})/6$. Up to a positive constant, $M = s/(1-s)$ and $Mw \to -1$. This scalar counterexample shows why terminal control is necessary. It makes no claim about a Newtonian trajectory.

9.4 Amplitude bounds and the terminal face

The two centrifugal contributions to the side-gap accelerations, after multiplication by R^2, are

$$Z\big((1-\eta)^2/m^2 - x^{-3}\big), \qquad Z\big(x^{-3} - n^2\eta^2/(m^2y^3)\big), \qquad Z = W^2.$$

Thus shape and the two history variables (Z, η) determine both terms. At a contact $\eta = h$, put $X_c = \delta x_s/W$, $Y_c = \delta y_s/W$, $S_c = h_x X_c + h_y Y_c$, and $P_c = m\delta^2(y^{-3} - 1)(k - h) < 0$. Direct differentiation gives

$$W\delta(\eta - h)_s = P_c - ZS_c.$$

Contacts with $S_c \geq 0$ point into $\eta < h$. On $S_c < 0$, the sufficient amplitude bound is $Z < Z_J := P_c/S_c$.

At an ordered syzygy with body 3 between bodies 1 and 2, write $y = q$, $x = 1 - q$. The launch angular-momentum signs require $0 < q < n/(m+n)$, and the collinear momentum constraints give

$$h = \eta_{\mathrm{col}} = \frac{q(m+1-q)}{(1-q)(n+q)}.$$

Canceling the common area-squared factor in P_c/S_c defines the terminal threshold Z_J^{syz}. Exact Bernstein certificates prove $1 < Z_J^{\mathrm{syz}} < Z_2$, where $Z_2 = R^2(\mathfrak{g}_{23} - \mathfrak{g}_{31})/[n^2\eta_{\mathrm{col}}^2/(m^2q^3) - (1-q)^{-3}]$ is the amplitude at which the second-gap acceleration vanishes, and $\mathfrak{g}_{ij} = \ddot{r}_{ij} - \ell_{ij}^2/r_{ij}^3$. The denominator is positive on this face. At the equal-leg endpoint the first-syzygy certificate gives $Z_* < 1$; its tied tangent satisfies $\partial_v[q(m+n)/n] > 18$. These bounds put nearby $v < 1$ members strictly inside the torque-compatible face with $Z < 1$. For the full family, neither interior noncontact nor the terminal amplitude bound is proved.

Energy and angular momentum alone do not imply the required bounds. For masses $(m, n, 1) = (4/5, 3/5, 1)$, consider an ordered collinear state with

$$(r_{12}, r_{23}, r_{31}) = (1, 697/700, 3/700), \qquad \eta = 419/32759, \qquad \ell_{23}^2 = \frac{476958362267}{2867548600}.$$

Choose zero radial velocities and the transverse velocities fixed by these angular momenta and $P = L = 0$. Exact arithmetic gives $H = -U_0$, pair angular-momentum signs $(-, +, -)$, and first-crossing area orientation $\dot{\Delta}_2/\ell_{23} = -292607/131036 < 0$, but

$$(r_{23} - r_{31})'' = -\frac{96354167469624287}{1827657180810} < 0.$$

This state is not asserted reachable from the tied initial brake. It shows that energy, collinear geometry, crossing direction, and pair angular-momentum signs do not suffice to control the second distance gap. The main analytic target is therefore a bound on *reachable histories*.

10 The remaining global problem

The stronger real-parameter conjecture would follow by excluding every positive zero of the brake residual for all $0 < u \leq \sqrt{2} - 1$. On a regular strict-maximum branch, the implicit-function theorem gives an event time $t_k(u)$, and the remaining condition is

$$\zeta(u, t_k(u)) \neq 0.$$

An argument restricted to regular branches must also account for their creation and termination, degenerate events, collisions, escape, and possible unbounded encounter counts. At a Jacobi degeneracy the Hopf residual replaces the complex chart. None of these obligations follows from counting equations and parameters.

The all-time results cover a narrow middle interval, an existential near-isosceles neighborhood, and infinitely many thin-triangle windows. First-turn exclusion and the planar collision neighborhoods further restrict the returning region, but do not control every later encounter. The torque equations isolate a possible analytic route through a bound on histories reached from the initial brake; the singular terminal estimate remains open. In the Jacobi–Maupertuis formulation, a second brake would be a geodesic chord between points of the degenerate boundary of $2(U-U_0)ds_m^2$, where $ds_m^2 = \sum_i m_i|dq_i|^2$ in the center-of-mass frame. No convexity or variational obstruction for the tied curve is yet known. If a real periodic point is found instead, the integer conjecture requires an additional exact arithmetic step. A real zero need not have rational u. Conversely, a countable dense set of rational parameters cannot be dismissed by a generic dimension count, a measure-zero claim, or a finite scan. The unresolved arithmetic question concerns exact intersections with the rational parameter curve.

A Proof dependencies and reproducibility

This manuscript incorporates the principal reductions, endpoint arguments, and local collision results from the earlier technical companion. The repository notes below supply complete polynomial certificates, verifier sources, assumptions, and archived outputs. The CAPD certificates use version 6.1.0 at commit `731079217a9254ea2948d742df2b170895effe7f`, with outward-rounded binary64 or MPFR arithmetic as specified in each record. The manuscript review checks the proof assembly and exact algebra and audits the archived middle-interval cover; it does not independently rerun every interval integration. The former companion remains an expanded research archive.

Result or ingredient	Repository proof/certificate entry point
Second-brake and maximum reduction	`docs/FABLE_EVENT_REDUCTION.md`
Terminal escape	`docs/ESCAPE_CRITERIA.md`; `docs/FABLE_MIDDLE_ESCAPE.md`
First maximum, $[.29, .29002]$	`docs/MIDDLE_FIRST_MAXIMUM_INTERVAL.md`
All-time middle interval	`docs/FABLE_MIDDLE_INTERVAL_NONPERIODICITY.md`
Near-equal-leg interval	`docs/ISOSCELES_SYZYGY_THRESHOLD.md`; `docs/ENDPOINTS.md`
First thin-triangle miss	`docs/SKINNY_REGULARIZATION.md`
Incoming-state matching	`docs/INCOMING_TAIL.md`
Parabolic transversality and scattering	`docs/COMPUTER_ASSISTED_TRANSVERSALITY.md`; `docs/COMPUTER_ASSISTED_TRANSVERSE_SCATTERING.md`
Explicit positive-density family	`docs/EXPLICIT_SKINNY_FAMILY.md`
First turns and returning trajectories	`docs/OUTGOING_ANGULAR_MOMENTUM.md`; `docs/SECOND_ENCOUNTER_MAP.md`
Triple endpoint and joint scaling	`docs/TRIPLE_ENDPOINT_SELECTION.md`; `docs/FINITE_B_TRIPLE_JOINT_BLOWUP.md`
Planar collision certificates	`docs/COMPUTER_ASSISTED_PLANAR_LIGHT_COLLISION_ROOT.md`; `docs/COMPUTER_ASSISTED_PLANAR_THIRD_FOURTH_COLLISION_ROOTS.md`
Finite-mass transfer and real collisions	`docs/DIFFERENTIATED_JOINT_INCLINATION.md`; `docs/REAL_COLLISION_SAMPLING.md`
Torque identities and correction	`docs/EXACT_REDUCTIONS.md`; `docs/FORCED_LAG_ENDPOINT_CORRECTION.md`
Counterexample search	`docs/PERIODIC_CONSTRUCTION_REPORT_2026-09-05.md`
Historical sources	`docs/PYTHAGOREAN_HISTORY_SOURCES.md`

The original fan image is reproduced without changing its pixels; its metadata are archived in `paper/figures/fan-provenance.json`. The script `paper/make_figures.py` generates the geometry figure and the archived defect plot from the exact parameterization and campaign data. The F_{30} trajectory image is also reproduced unchanged; its separate plotting audit is `paper/figures/f30-plot-audit.txt`. Build instructions, figure provenance, and the preserved pre-review source and PDF are described in `paper/README.md`.

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